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LOCALIZED ROTATION DURING PALEOGENE EXTENSION IN EAST CENTRAL IDAHO: PALEOMAGNETIC AND GEOLOGIC EVIDENCE

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Abstract. We collected paleomagnetic data from 137 sites in middle Eocene volcanic rocks of the northeastern Basin and Range province in order to characterize the three-dimensional kinematics of continental extension in the Lost River and Lemhi ranges, Idaho, and to test previous block rotation models. Statistically significant flattening of directions at three of 10 localities reflects either insufficient averaging of paleosecular variation or erroneous tilt corrections. Five of the seven remaining locality means are statistically indistinguishable from the middle Eocene reference direction. This indicates that vertical axis rotation did not accompany post-50 Ma normal faulting. Discordant declinations from two localities in the northern Lost River Range indicate 30°-40° of counterclockwise rotation of a large domain of NNW striking volcanic rocks. Inclination flattening at one of the two rotated localities is moderate but statistically significant ($F \pm \Delta F = 9.9° \pm 6.5°$). In total, the paleomagnetic data conflict with earlier predictions of uniform clockwise rotation in response to right lateral transtension adjacent to the eastern Snake River Plain. There is a close correspondence between counterclockwise rotated localities and domains of NNW striking volcanic rocks and between unrotated localities and domains of north striking volcanic rocks. New mapping shows that counterclockwise rotated localities are all in the hanging wall of a major west dipping low-angle normal fault, the Donkey fault (named for the first time). This fault loses displacement southwards, was active in Eocene-Oligocene time and is probably a tilted high angle fault. The spatial distribution of rotated and unrotated localities is best explained by counterclockwise rotation of the hanging wall of the Donkey fault during opening of a southward tapering half graben. According to this hanging wall rotation model the footwall of the Donkey fault remained fixed relative to its rotating hanging wall. Our proposed kinematic model is generally applicable to extending continental lithosphere and provides a mechanism for coeval crustal thinning and vertical axis rotation. Thus vertical axis rotation may occur in the absence of strike-slip faults in extensional orogens.

INTRODUCTION

The possibility of significant vertical axis rotation during crustal extension only recently has been addressed using paleomagnetic and geological means [Ron et al.,1984; Hudson and Geissman,1987; Nelson and Jones, 1987; Hagstrum and Gans, 1989; Wells and Heller, 1988; Wells and Hillhouse, 1989; Li et al., 1990; Calderone et al., 1990]. It is not clear whether block rotations characterize most extended regions or only regions crosscut by strike-slip faults [e.g., Nelson and Jones, 1987]. Block rotation may preferentially arise in highly extended terrains [e.g., Wells and Hillhouse, 1989] but not in moderately extended regions [e.g., Calderone et al., 1990]. Geometric and kinematic models of block rotation typically predict uniform sense and amount of rotation in response to slip on parallel fault sets [e.g., Freund, 1970; McKenzie and Jackson, 1983, 1986; Garfunkel and Ron, 1985; Ron et al., 1984], yet paleomagnetic studies indicate that rotations may be quite heterogeneous in extended regions [e.g., Wells and Hillhouse, 1989; Hagstrum and Gans, 1989; Nelson and Jones, 1987]. A clearer understanding of the three-dimensional kinematics of continental extension is the ultimate goal of our integrated structural and paleomagnetic study of Cenozoic deformation in east central Idaho. Our approach to this problem has been twofold. We have (1) investigated possible vertical axis rotations using

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paleomagnetic data from the Eocene Challis Group volcanic rocks and (2) collected kinematic data along extensional structures in the area to assess independently proportions of strike slip and dip slip on Cenozoic extensional faults [Janecke, 1989a]. Determining the areal extent, magnitude, and sense of vertical axis rotation is a critical component of our structural and tectonic study. This information is used to determine the Paleogene paleostress field of east central Idaho [Janecke, 1989a]. Throughout this paper, rotation refers to rigid body rotation about vertical axes, and tilt refers to rigid body rotation about horizontal or subhorizontal axes.

This paper presents results of paleomagnetic analysis of 10 localities, comprising 137 sites (1026 samples), in middle Eocene Challis volcanic rocks. High-quality data from key localities, when combined with dip domain patterns of the volcanic rocks, reveal that (1) half of our localities are unrotated and (2) of those localities which show evidence for vertical axis rotation, the majority record counterclockwise rotation. Counterclockwise rotation affected the entire northern Lost River Range. The paleomagnetic results do not support the predictions of a model by McKenzie and Jackson [1986] for a period of uniform clockwise rotation in the region in response to overall right-lateral transtension. A large domain of counterclockwise rotation in the northern Lost River Range may record differential counterclockwise rotation of the hanging wall of a major low-angle normal fault (herein referred to as the Donkey fault) relative to its footwall (Figure 1).

BACKGROUND AND GEOLOGICAL SETTING

Structural Geology

The study area is located in the northeastern corner of the Basin and Range province (Figure 1), an area that is actively extending along moderately WSW dipping normal faults. Basin and Range topography is well developed, and the western flanks of the Lost River, Lemhi, and Beaverhead ranges are marked by Pleistocene to Holocene fault scarps [Scott et al., 1985; Haller, 1988; Crone and Haller, 1989]. The region is situated within the Intermontane Seismic Belt, and was the site of the M_S 7.3 Borah Peak earthquake [Doser and Smith, 1985].

Active Basin and Range normal faults overprint two earlier extensional fault systems in the Lost River and Lemhi ranges [Janecke, 1989a]. The earliest normal faults extended east central Idaho in a NW-SE direction and were active during Eocene time [Baldwin , 1951; McIntyre et al., 1982; Wust, 1986; O'Neill and Pavlis, 1988; Janecke, 1989a, 1989b]. Unroofing of the Pioneer core complex (Figure 1), slip on the Wildhorse detachment fault, displacement on associated hanging wall faults in the study area, and formation of the Trans-Challis fault zone appear to be roughly synchronous with volcanism in the middle Eocene Challis volcanic field [McIntyre et al., 1982; McIntyre and Johnson, 1985; Ekren, 1985; Fisher, 1985; Skipp and Harding, 1985; Kiilsgaard et al., 1986; Bennett, 1986; Wust, 1986; O'Neill and Pavlis, 1988]. New $^{40}Ar/^{39}Ar$ ages of 32-36 Ma (latest Eocene to earliest Oligocene) on the mylonites in the Pioneer core complex may indicate that some WNW directed extension postdated middle Eocene Challis magmatism [Silverberg, 1988].

A second phase of extension began about 45-46 Ma (middle Eocene) in east central Idaho and continued into Oligocene time [Janecke, 1989b; Janecke and Snee, 1990]. A system of large-offset west directed low-angle normal faults [Hait, 1984; Janecke, 1989b] crosscuts the NE striking fault system and probably soled at depth into a regional west dipping detachment fault. Thick sections of coarse syntectonic conglomerates were shed into extensional half grabens during this second period of regional extension. Low-angle normal faults in the study area dating from this phase of extension include (from SE to NW); the Pass Creek fault system, the Sawmill Canyon fault, and the Donkey fault (Figure 1). The high-angle Allison Creek fault (Figure 1) also dates from this time.

The Donkey fault slices NNW across the Donkey Hills and projects northward toward the Lemhi Range. The active Basin and Range fault along the west flank of the Lemhi Range truncates the Donkey fault. At its southern end the Donkey fault extends into the Lost River Range, losing displacement toward the south, and eventually becomes impossible to trace. Near its northernmost exposed end in the northern Donkey Hills the fault omits a 2-3 km thick section of Challis volcanic rocks, the entire miogeoclinal Paleozoic section (about 7 km thick), and the top of the Precambrian section, juxtaposing tilted Tertiary gravels against Proterozoic quartzites. This corresponds to a minimum of 13 km of dip slip. At the southern mapped trace of the fault, Eocene volcanic rocks comprise both the footwall and hanging wall of the fault. This fault is a tilted high angle normal fault.

Paleomagnetic sampling localities at Meadow Creek, Spring Hill, Garrett Rock, and Long Creek are within (or can be interpreted to lie within) the hanging wall of the Donkey fault (Figure 1). The Copper Lake and Massacre Mountain localities lie within the hanging wall near the southern end of the fault. All but one of the rotated localities are located within the hanging wall of the Donkey fault.

Geological and Geophysical Evidence Suggestive of Vertical Axis Rotation

Our evaluation of vertical axis rotation in the Lost River and Lemhi ranges in east central Idaho was prompted by a variety of geological and geophysical observations suggestive of Cenozoic block rotations as well as recent kinematic modeling [McKenzie and Jackson, 1986]. Previous mapping [Ross, 1947; Ruppel and Lopez, 1981, 1988; Mapel et al., 1965; Mapel and Shropshire, 1973; McIntyre and Hobbs, 1987] showed that belts of the middle Eocene Challis volcanic rocks in the study area fall into discrete domains, each characterized by a uniform dip direction and roughly uniform dip magnitude (Figure 2). Some of the dip domains are regionally extensive, while others characterize small blocks with maximum dimensions of the order of 5 to 10 km. Pronounced strike changes between domains suggested to us that vertical axis rotation was likely in the study area.

In addition, geologic mapping indicates that subhorizontal Mesozoic fold axes are not parallel throughout the region (Figure 2). Folds trend roughly north to NNW in the east central and southern Lost River Range and central and southern Lemhi Range but swing to more NNW to northwesterly trends in the central and northern Lost River Range [Mapel et al., 1965; Mapel and Shropshire, 1973; Holmes and Evans, 1989; Beutner, 1968; Hait, 1965; Ross, 1947; Ruppel and Lopez, 1981, S. U. Janecke, unpublished mapping, 1987-1989; B. Skipp, unpublished mapping, 1978]. Differential vertical axis rotation is one possible explanation for the contrasting fold trends in the two areas. The changes in trend between the northern and southern areas are typically 15° to 40°.

Seismological, geodetic, and geological evidence for oblique slip on the Lost River fault during the 1983 Borah Peak earthquake has been interpreted to indicate block rotations

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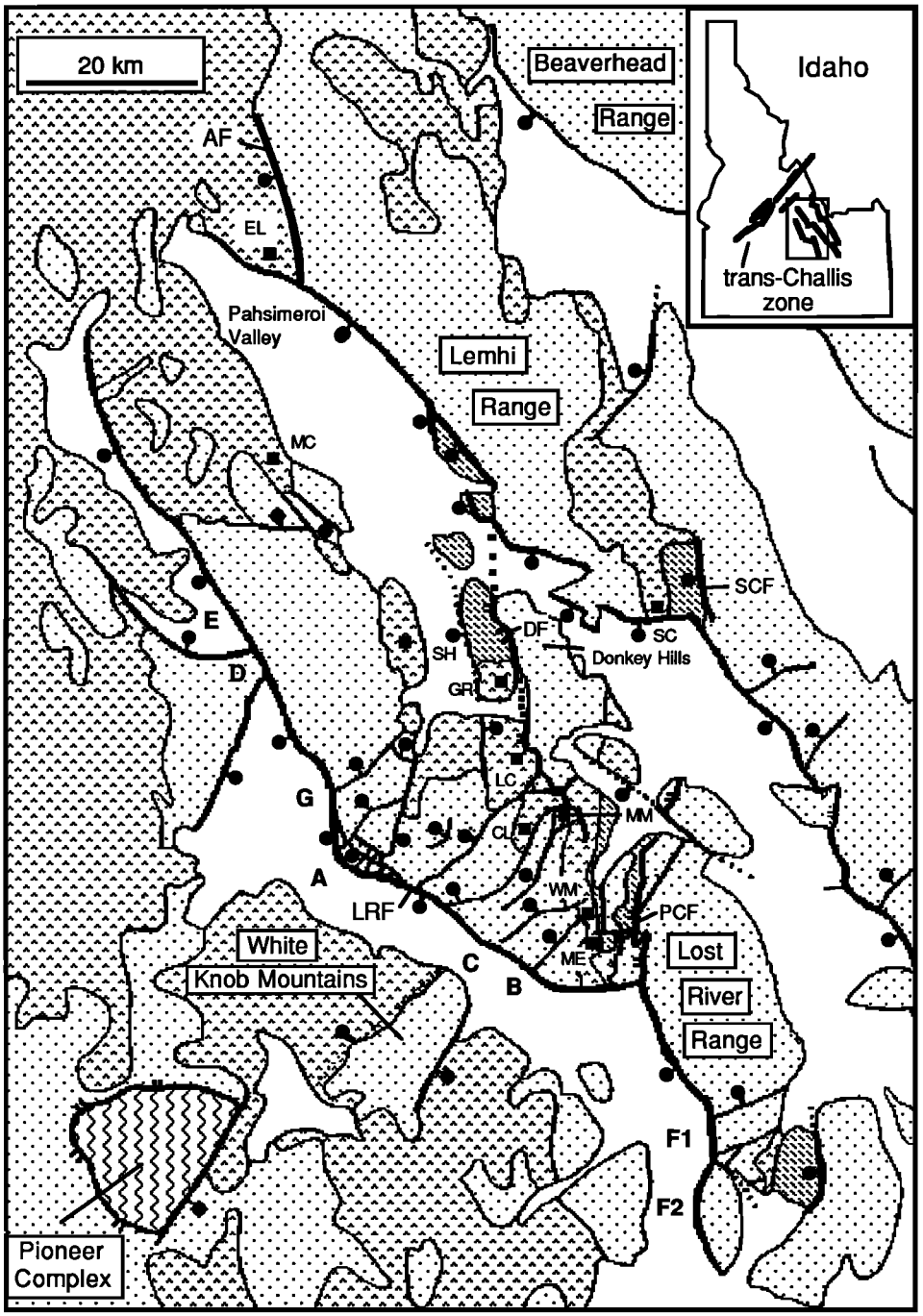


Fig. 1. Simplified geologic map of the study area in east central Idaho. Localities are denoted by small solid squares. Abbreviations are CL, Copper Lake; EL, Ellis; GR, Garrett Rock; LC, Long Creek; MM, Massacre Mountain; MC, Meadow Creek; ME, Methodist Creek; SC, Sawmill Canyon; SH, Spring Hill; WM, Warren Mountain; LRF, Lost River fault; PCF, Pass Creek fault; AF, Allison Creek fault; SCF, Sawmill Canyon fault; and DF, Donkey fault. Locations along the Lost River fault where slip estimates were made are indicated by letters A-G. Geology modified from Bond [1978] and S.U. Janecke (unpublished mapping, 1987-1989). Patterns are inverted v, middle Eocene Challis strata; dots, Precambrian and Paleozoic strata; dashes, Paleogene sedimentary rocks; wavy lines, footwall of Pioneer core complex; unpatterned areas are Late Cenozoic basin-fill. Bar and ball on downthrown side of moderate to high-angle normal faults, dotted where concealed. Double bar symbol on downthrown side of low-angle normal faults.

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Fig. 2. Dip domain map of the Challis volcanic rocks (dotted pattern) showing mostly easterly dips in the central and southern part of the area and northeasterly dips in the northern Lost River Range. Fold axes, shown as dashed lines, have northerly trends in the south and northwesterly trends in the north. Fold axes from Rember and Bennett [1979a, 1979b and references therein], Fisher et al. [1983], Holmes and Evans [1989], B. Skipp (unpublished mapping of Sunset Peak quadrangle, 1978), and S.U. Janecke (unpublished mapping, 1987-1989). Refer to Figure 1 for labels and symbols.

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[McKenzie and Jackson, 1986; Westaway, 1989]. A 36 km long section of the Lost River fault ruptured as a oblique-normal left-lateral fault during the M_S 7.3 earthquake [Crone and Machette, 1984] (seismological data summarized by Richins et al. [1987] and Crone et al. [1987]. The oblique slip noted in the focal mechanism solutions and surficial data prompted McKenzie and Jackson [1986] to apply a transtensional block rotation model to east central Idaho. In their model, left-normal slip along the NNW striking normal faults in the region is viewed as a response to right-lateral transtension within a regional NE striking zone. Two rigid plates move past and away from each other across the right-lateral zone (Figure 3), and fault-bounded crustal blocks within the deforming zone rotate in a clockwise sense. The McKenzie and Jackson model predicts homogeneous clockwise rotation of blocks comprising the Lost River, Lemhi, and Beaverhead ranges in late Cenozoic time.

Challis Volcanic Rocks

The middle Eocene Challis volcanic rocks are the focus of paleomagnetic investigation and structural analysis. These volcanic rocks completely blanketed eastern and central Idaho in middle Eocene time [McIntyre et al., 1982; McIntyre, 1985; Snider and Moye, 1989]. This thick sequence of lava flows and tuffs postdates intense Sevier folding and thrusting [Ross, 1947; Skipp and Hait, 1977; Mapel and Shropshire, 1973] and predates most of the extensional deformation [Baldwin, 1951; Ross, 1947; Janecke, 1989a]. Volcanic rocks rest unconformably on a high-relief erosion surface developed on upper Paleozoic carbonate strata. Volcanic rocks in the study area consist primarily of lava flows of intermediate composition, minor interbedded airfall and ash flow tuffs, volcaniclastic sedimentary rocks, and small dikes, sills, and plugs. The study area is SE and east of the major caldera complexes of the Challis volcanic field [McIntyre et al., 1982; Snider et al., 1989; Snider and Moye, 1989], so tilting and rotation due to caldera collapse is unlikely. Airfall and ash flow tuffs at the top of the section tend to be thin, unwelded to weakly welded, and were probably distal to vent areas. Lava flows from local vent areas comprise the bulk of the 1.5-3 km thick section of volcanic rocks in the central Lost River and Lemhi ranges [Mapel and Shropshire, 1973; Ruppel and Lopez, 1981; McIntyre and Hobbs, 1987; S. U. Janecke, unpublished data, 1987-1989]. Thick sequences are exposed at several localities in the study area, affording numerous excellent opportunities for paleomagnetic sampling.

New ^{40}Ar-^{39}Ar isotopic age determinations from the main part of the Challis volcanic rocks in the Lost River and Lemhi ranges span about 1.5 m.y. between 49.3 and 47.7 Ma [S. U. Janecke and L. W. Snee, unpublished data,1989) Our age determinations are from lava flows of intermediate composition and interbedded tuffs. All but one of the units sampled paleomagnetically appear to be older than 48 Ma. Some paleomagnetic samples were collected stratigraphically beneath the oldest dated flows. K-Ar and ^{40}Ar-^{39}Ar isotopic age determinations on similar rocks in nearby ranges suggest that the entire Challis sequence erupted between 51 and 45 Ma [Armstrong, 1975; McIntyre et al., 1982; Snider et al., 1989; Snider and Moye, 1989].

METHODS AND SAMPLING

Our sampling program was designed to assess possible vertical axis rotations in each of the major dip domains of the Challis sequence, as well as to compare rotations in both

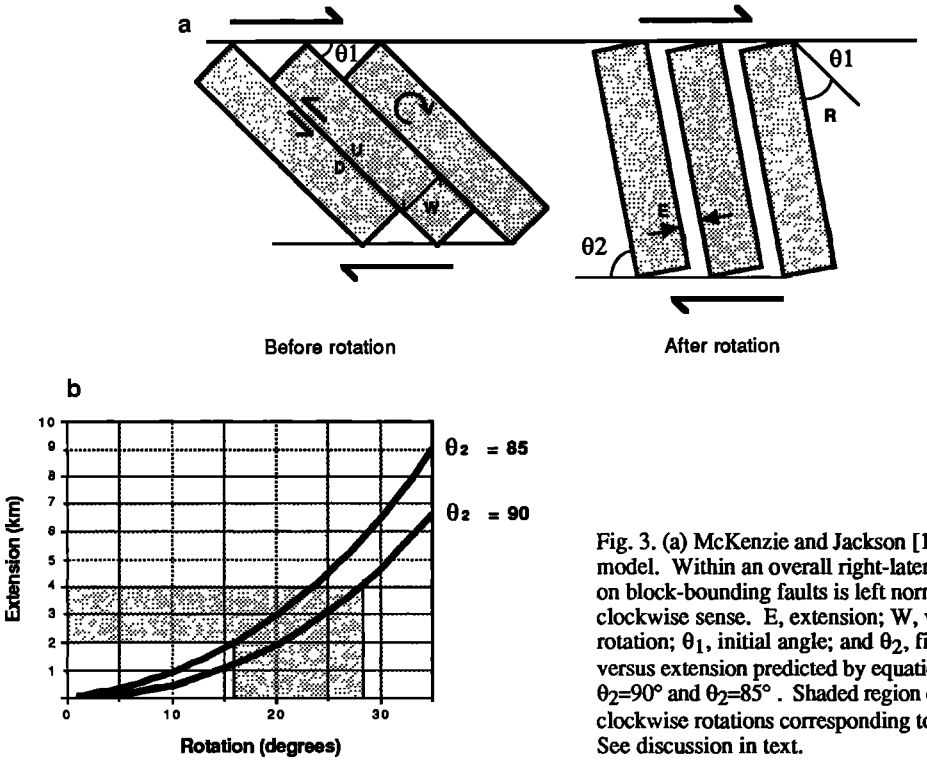


Fig. 3. (a) McKenzie and Jackson [1983, 1986] block rotation model. Within an overall right-lateral transtensional zone, slip on block-bounding faults is left normal, and blocks rotate in a clockwise sense. E, extension; W, width of block; R, rotation; θ_1, initial angle; and θ_2, final angle. (b) Rotation versus extension predicted by equation (1) for W= 30 km with θ_2=90° and θ_2=85° . Shaded region denotes range of expected clockwise rotations corresponding to fault heaves of 2-4 km. See discussion in text.

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steeply tilted and gently tilted regions (Figures 1 and 2). Two localities (Copper Lake (CL) and Massacre Mountain (MM)) were sampled specifically to assist in correlating lava flows and tuffs across different structural blocks. Seven localities are in the central Lost River and Lemhi ranges, where detailed geologic mapping and structural analysis best constrain the geometry of the volcanic units. For completeness, we also sampled three localities outside of the structural study area (Ellis (EL), Meadow Creek (MC), and Spring Hill (SH) localities) in regions where the history of events is less well known. The volcanic rocks in these areas are typically less tilted than sequences in the central region.

The Methodist Creek, Warren Mountain, Massacre Mountain, and Sawmill Canyon localities occur in a large east dipping domain of Challis volcanic rocks extending north from the Pass Creek area of the Lost River Range into the Sawmill Canyon area in the Lemhi Range (Figures 1 and 2). The Spring Hill and Meadow Creek localities are within a large domain of NE to ENE dipping volcanic rocks in the northern Lost River Range. The Ellis locality (EL) in the northern Lemhi Range (Figure 1) was collected from subhorizontal flows resting on the 46.4±1.6 Ma tuff of Ellis Creek [McIntyre et al., 1982]. The remaining localities (Garrett Rock, Long Creek, and Copper Lake) are situated in small fault blocks in the central Lost River Range. This area is structurally complex, and dip directions vary from block to block. Volcanic sequences dip toward the southeast at the Garrett Rock locality and roughly toward the east at the Long Creek and Copper Lake localities.

Samples for paleomagnetic analysis were collected at 10 localities comprising 148 sites. We report results from 137 of these sites in Table 1. Samples from 11 sites exhibited extremely unstable demagnetization behavior, in many cases recording both normal and reversed polarities within a single site, and were rejected outright. Each site represents an individual lava flow, cooling unit within an ash flow tuff, or small intrusive body (dike or sill). The vast majority of sites were collected in lava flows. Widespread glacial and periglacial activity during Pleistocene time resulted in thick colluvial, alluvial, and morainal cover on many hill slopes. Good exposures are, unfortunately, restricted to ridge tops where frequent lightning strikes complicate paleomagnetic investigations.

Every site at an individual locality is within a single uniformly tilted fault block; a single structural correction was therefore applied to every site at a locality. We sampled a few intermediate to mafic plugs, sills, and dikes. These appear to be roughly coeval with their volcanic host rocks. Flows of similar composition are interlayered in the sequence and may have been fed by these intrusions; there is almost no evidence for younger intermediate to mafic magmatism in this area. The same tilt correction was therefore applied to the intrusive rocks as to the enclosing volcanic rocks. No systematic differences between site mean directions of paleomagnetism from extrusive and intrusive rocks were observed.

The attitudes of flow sequences in some localities are difficult to determine because exposures are discontinuous and well-exposed sedimentary and/or tuffaceous interbeds are uncommon within the lava pile. Where possible, attitudes of interbedded sedimentary strata or compaction foliation in ash flow tuffs were measured. In some sections we resorted to using laminations of airfall tuffs for determination of attitude. Strikes and dips of bedded volcanic sequences, three-point solutions for attitudes of individual lava flows or tuffs within the sequence, and sightings of the overall strike and dip of a package of tilted lava flows provided additional constraints on the structural correction. Studies in the Challis volcanic rocks NW and west of the Lost River Range indicate relatively subdued, plateau-like topography during volcanism after initial infilling of inherited topography [Snider and Moye, 1989]. Evidence for steep-sided stratovolcanoes is lacking, and magmas apparently erupted from fissure vents. Thus airfall tuffs and stacked sequences of lava flows are probably fairly good indicators of paleohorizontal. If lava flows were originally deposited on an inclined surface, however, some structural corrections could be in error. We selected sampling localities judiciously in order to minimize these problems, but they could not be eliminated completely.

Samples were collected with a portable gas drill. Eight or more 2.5 cm diameter cores were collected at each site and oriented with both magnetic and sun compasses. Remanent magnetizations were measured by a three-axis cryogenic magnetometer (2G Enterprise 760 R) in a magnetically shielded space. Alternating field (AF) demagnetizations of samples collected in 1987 were carried out using a standard manual AF demagnetizer, while samples from our 1988 and 1989 collections were demagnetized with an automated sample handling and AF demagnetization system. Detailed pilot AF demagnetization analyses of one sample at each site (10 to 22 steps) were used to design a demagnetization program for the remainder of samples at the site. Five to 12 alternating field demagnetization steps were performed on the majority of the samples (Figure 4). Thermal demagnetization of representative specimens shows that for most lithologies the thermally unblocked characteristic remanence direction was identical to that isolated during alternating field demagnetization (e.g., Figures 4c and 4f).

Well-defined and well-grouped components of magnetization were isolated during progressive demagnetization of most sites. Linear trajectories of vector endpoints toward the origin (Figure 4) suggest that a stable characteristic remanence component was isolated. Other sites behaved in a less straightforward fashion and required remagnetization circles analysis, especially samples with high NRM intensity, which presumably came from lightning-affected sites (e.g., Figure 4m). Sites with poor demagnetization behavior were rejected and are not reported in Table 1. Fine-grained magnetite appears to be the principal carrier of remanence in these rocks. We interpret the characteristic magnetization of each flow or intrusion as a primary thermoremanent magnetization acquired by the flows, tuffs, and shallow intrusions during original cooling.

All sites with cones of 95% confidence (α_{95}) greater than our arbitrary cutoff of 18° were excluded. This resulted in the rejection of two sites from the Massacre Mountain locality. In addition, we rejected sites with a virtual geomagnetic pole (VGP) more than 38° from their respective locality mean poles, following Diehl et al. [1983]. Three sites at Spring Hill and one site at Methodist Creek were eliminated by this test.

We used the 44-54 Ma paleomagnetic pole of Diehl et al. [1983] to calculate an expected direction for east central Idaho. This pole has a latitude and longitude of 82.8° N, 170.4° E and an A_{95} of 3°. The pole is a 10 m.y. average of 147 sites centered on the time span represented by extrusion of Challis Group rocks. We consider this pole to be quite reliable because of the large number of independent data points used in its determination and because of the near stillstand of the pole relative to North America at this time [Diehl et al., 1983].

Localities in our study span 0.7° of latitude and 0.6° of longitude. The inclination of the reference direction at the northern locality is about 0.6° steeper than that of the southernmost locality. The expected declination at each

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TABLE 1. Site Mean Directions and Statistics

Site	Dec	Inc	α_{95}	k	CSD	R	N/N_o
			Copper Lake				
CL81	326.4	35.1	3.6	288.6	4.8	6.9792	7/7
CL82	326.5	37.4	6.4	74.8	9.4	7.9064	8/8
CL83	329.6	38.7	4.3	199.0	5.7	6.9700	7/8
CL84*	334.0	44.2	6.5	126.0	5.1		7/8
CL85	131.1	-21.8	9.3	36.6	13.4	7.8088	8/8
CL86	129.4	-41.3	5.6	184.9	5.9	4.9794	5/7
CL87	112.2	-19.6	6.9	75.7	9.3	6.9208	7/8
CL88	102.4	-42.2	12.2	25.5	16.0	6.7647	7/8
CL89	80.2	-33.9	9.3	43.2	12.3	6.8610	7/8
CL90	114.9	-37.4	6.4	112.2	7.6	5.9554	6/7
CL91	118.0	-41.0	8.0	57.4	10.7	6.8955	7/7
CL92	109.8	-29.3	3.6	284.8	4.8	6.9789	7/8
			Ellis Creek				
EL115	174.6	-50.0	6.5	88.2	8.6	6.9320	7/7
EL116	125.6	-64.9	4.6	172.6	6.2	6.9652	7/7
EL117	172.8	-60.0	3.9	237.4	5.3	6.9747	7/8
EL118	198.9	-66.8	4.0	233.7	5.3	6.9743	7/8
EL119*	151.0	-50.7	4.8	493.5	2.6		5/8
EL120	162.2	-60.5	5.5	149.2	6.6	5.9665	6/8
EL121	196.3	-61.6	7.2	71.3	9.6	6.9158	7/8
			Garrett Rock				
GR93	154.5	-21.7	5.6	118.4	7.4	6.9493	7/7
GR94	169.2	-22.4	9.1	71.9	9.5	4.9444	5/7
GR95	168.7	-22.3	5.2	311.8	4.6	3.9905	4/7
GR96	173.8	-12.3	6.9	96.1	8.3	5.9481	6/8
GR97	161.3	-29.6	10.3	55.8	10.8	4.9283	5/6
GR98	161.5	-13.7	7.2	70.8	9.6	6.9147	7/8
GR227	196.8	-28.8	3.2	307.7	4.6	7.9773	8/10
GR228	163.8	-38.2	10.5	34.3	13.8	6.8250	7/8
GR229*	141.0	-16.0	16.0	127.4			5/7
GR230*	149.9	-20.6	5.4	158.3	4.6		8/9
GR231	157.2	-19.8	4.7	690.4	3.1	2.9971	3/8
GR232	146.4	-20.6	3.1	316.0	4.6	7.9778	8/8
GR233	154.2	-16.3	8.2	55.3	10.9	6.8915	7/8
			Long Creek				
LC69	119.7	-6.6	5.4	128.2	7.1	6.9532	7/7
LC70	117.1	-7.5	9.8	48.1	11.7	5.8960	6/7
LC71	118.6	-15.1	5.3	132.0	7.0	6.9547	7/8
LC72	117.1	-4.5	9.6	40.6	12.7	6.8521	7/8
LC73	118.2	-7.1	6.5	86.7	8.7	6.9308	7/7
LC74	123.7	-12.5	2.8	471.5	3.7	6.9873	7/8
LC75	115.6	-3.3	5.7	114.0	7.6	6.9474	7/7
LC76	127.6	3.6	6.4	110.5	7.7	5.9548	6/7
LC77	115.7	-0.3	7.6	64.6	10.0	6.9071	7/7
LC78	118.4	-5.8	10.0	37.2	13.3	6.8389	7/8
LC79	110.5	-11.9	5.7	112.1	7.6	6.9464	7/7
LC80	104.7	8.7	6.1	98.7	8.1	6.9392	7/7
			Massacre Mountain				
MA99	315.8	30.0	6.7	101.0	8.1	5.9508	6/7
MA100	320.3	36.3	5.9	105.4	7.9	6.9431	7/8
MA101*†	289.3	23.5	67.9	99.4			7/7
MA102	125.7	-27.0	8.1	48.2	11.7	7.8548	8/8
MA103*	173.2	-35.6	11.5	47.5	8.3		7/9
MA104	311.1	34.7	7.8	61.4	10.3	6.9022	7/7
MA105	116.9	-19.0	3.3	331.3	4.4	6.9819	7/7

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Table 1. (continued)

Site	Dec	Inc	α_{95}	k	CSD	R	N/N_o
MA106*	302.8	45.1	12.1	53.1	7.9		6/7
MA107*	310.8	38.3	8.9	73.3	6.7		7/8
MA108*	149.7	-40.3	4.8	370.1			4/8
MA109*	173.5	-50.3	11.9	40.5	9.0		8/8
MA110*†	317.6	29.5	28.4	27.3			6/7
MA111	115.7	-23.2	4.7	267.9	4.9	4.9851	5/8
MA112*	106.3	-30.2	5.3	235.9	3.7		8/9
			Meadow Creek				
MC1	279.0	58.6	4.3	323.4	4.5	4.8976	5/5
MC2	296.4	59.6	5.9	104.8	7.9	6.9427	7/8
MC3	115.7	-48.6	5.1	176.4	6.1	5.9717	6/7
MC4	122.8	-47.4	4.2	206.4	5.6	6.9709	7/7
MC5	122.9	-48.0	3.7	323.1	4.5	5.9845	6/8
MC6	120.1	-48.6	4.1	272.6	4.9	5.9817	6/7
MC7	57.9	-60.7	14.0	24.0	16.6	5.7913	6/9
MC8	127.6	-43.7	2.7	502.5	3.6	6.9881	7/7
MC9	126.0	-42.4	3.5	295.9	4.7	6.9797	7/7
MC10	139.3	-43.8	7.6	63.8	10.1	6.9060	7/7
MC11	124.1	-48.6	5.1	580.8	3.4	2.9966	3/6
MC12†	79.7	-73.8	2.7	616.9	3.3	5.9919	6/7
			Methodist Creek				
ME46	133.4	-48.9	7.4	68.1	9.8	6.9119	7/7
ME47	142.1	-43.3	4.4	157.8	6.4	7.9556	8/8
ME48	139.5	-36.0	4.4	157.6	6.4	7.9556	8/8
ME49	138.0	-39.5	5.6	118.6	7.4	6.9494	7/7
ME50	122.0	-55.9	5.7	81.7	8.9	8.9020	9/9
ME51	121.1	-36.6	3.2	322.2	4.3	6.9830	7/7
ME52	124.5	-45.4	7.4	56.8	10.7	7.8787	8/8
ME53†	100.6	57.9	15.4	25.7	16.0	4.8445	5/8
ME54	129.0	-6.6	10.5	54.6	10.95	4.9268	5/9
			Sawmill Canyon				
SC13	297.6	46.9	6.5	73.5	9.4	7.9047	8/8
SC14	310.6	41.5	6.8	68.1	9.8	7.8972	8/8
SC15*	309.1	32.9	7.1	83.7	6.3		8/9
SC16	310.2	36.0	5.2	138.2	6.9	6.9566	7/8
SC17	289.3	46.7	9.2	54.2	11.0	5.9077	6/8
SC18*	299.5	43.8	9.2	65.4	7.1		7/9
SC20*	322.7	58.4	15.5	20.0	12.8		7/8
SC21	289.7	22.1	3.9	205.0	5.7	7.9658	8/8
SC22*	296.2	21.7	12.0	30.7	10.3		8/8
SC23	256.9	28.7	8.6	42.2	12.5	7.8341	8/8
SC24*	303.6	25.4	6.2	135.1	4.9		7/7
SC25	296.1	24.9	5.3	112.1	7.6	7.9376	8/8
SC142	300.3	29.9	12.2	31.4	14.5	5.8405	6/6
SC141	282.8	48.3	3.6	285.6	4.8	6.9790	7/7
SC140	118.0	-45.6	4.8	158.7	6.4	6.9622	7/7
SC139	119.1	-31.5	4.6	175.5	6.1	6.9658	7/7
SC138	124.9	-29.5	9.2	100.9	8.1	3.9703	4/8
SC137	103.4	-23.2	2.5	574.5	3.4	6.9896	7/7
SC136	91.3	-20.3	6.0	102.5	8.0	6.9415	7/7
SC135	132.1	-9.7	7.3	85.7	8.7	5.9417	6/7
SC134	154.2	-21.6	4.2	209.3	5.6	6.9716	7/7
SC133	289.7	37.4	2.0	882.0	2.7	6.9932	7/7
			Spring Hill				
SH35	97.2	-52.3	5.2	170.3	6.2	5.9706	6/7
SH36	97.0	-43.7	3.9	201.6	5.7	7.9653	8/8
SH37*	78.4	-39.1	17.8	38.1	9.3		6/8

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Table 1. (continued)

Site	Dec	Inc	α_{95}	k	CSD	R	N/N_o
			Spring Hill				
SH38	87.9	-46.1	2.5	567.9	3.4	6.9894	7/7
SH39	87.1	-47.2	3.7	223.2	5.4	7.9686	8/7
SH40	87.7	-48.7	4.0	234.6	5.3	6.9744	7/7
SH41	85.0	-44.5	2.2	759.1	2.9	6.9921	7/7
SH42	125.3	-13.4	3.0	393.4	4.1	6.9847	7/8
SH43	101.0	-31.6	7.8	61.2	10.4	6.9019	7/7
SH44	101.2	-30.2	6.0	75.3	9.3	8.8938	9/9
SH45	105.3	-16.6	10.4	54.9	10.9	4.9272	5/8
SH212	99.9	-28.4	2.9	432.0	3.9	6.9861	7/7
SH213	105.7	-17.2	9.4	42.6	12.4	6.8591	7/7
SH214*†	111.6	33.0	12.0	54.4	7.8		8/8
SH215*†	88.0	17.8	6.7	95.5	5.9		8/8
SH216*†	100.0	63.7	46.4	11.7			7/7
SH217	104.7	-31.4	13.4	33.8	13.9	4.8816	5/7
SH218	108.4	-24.8	3.2	368.9	4.2	6.9837	7/8
SH219*	124.7	-51.6	10.6	63.8	7.2		6/7
SH220*	94.2	-30.0	10.3	40.0	9.1		8/8
SH221	92.3	-40.2	5.7	257.6	5.0	3.9884	4/7
SH222*	111.2	-21.0	14.0	60.6	7.4		5/7
SH223*	109.1	-40.3	5.7	370.9	3.0		5/7
SH224*	102.5	-18.8	7.7	83.5	6.3		9/10
SH225	83.5	-12.0	8.9	106.7	7.8	3.9719	4/7
SH226	114.0	-38.9	7.6	10.3	8.0	4.9612	5/7
			Warren Mountain				
WM60	197.3	-77.7	8.2	127.8	7.2	3.9765	4/8
WM61	140.3	-60.1	4.9	129.5	7.1	7.9459	8/8
WM62	143.9	-41.4	3.9	201.2	5.7	7.9652	8/10
WM63	161.6	-37.1	9.0	39.0	13.0	7.8203	8/8
WM64	145.0	-49.7	5.5	151.9	6.6	5.9671	6/7
WM65	186.6	-51.9	3.8	249.1	5.1	6.9759	7/8
WM66	147.8	-50.0	6.1	158.9	6.1	4.9748	5/8
WM67	131.0	-48.2	2.8	452.9	3.8	6.9868	7/7
WM68	122.4	-45.7	4.5	184.2	5.9	6.9674	7/7

Dec and Inc are declination and inclination of site mean in degrees; α_{95} is radius of 95% confidence circle in degrees; k is estimated Fischer precision parameter for site mean magnetization parameter; CSD is circular standard deviation; R is statistical parameter of Fisher [1953]; and N/N_o is number of samples included in statistical analysis/ number of samples analyzed.

* Analysis by remagnetization circles or combined remagnetization circles and stable endpoints.

† Excluded from locality mean calculation (see text for explanation).

locality is essentially the same. We thus calculated the reference declination and inclination for each individual locality and used it to determine the rotation and flattening values and errors for each locality following Beck [1980] and Demarest [1983]. The expected declination is 350° and the expected inclination ranges from 63.7° to 64.3°.

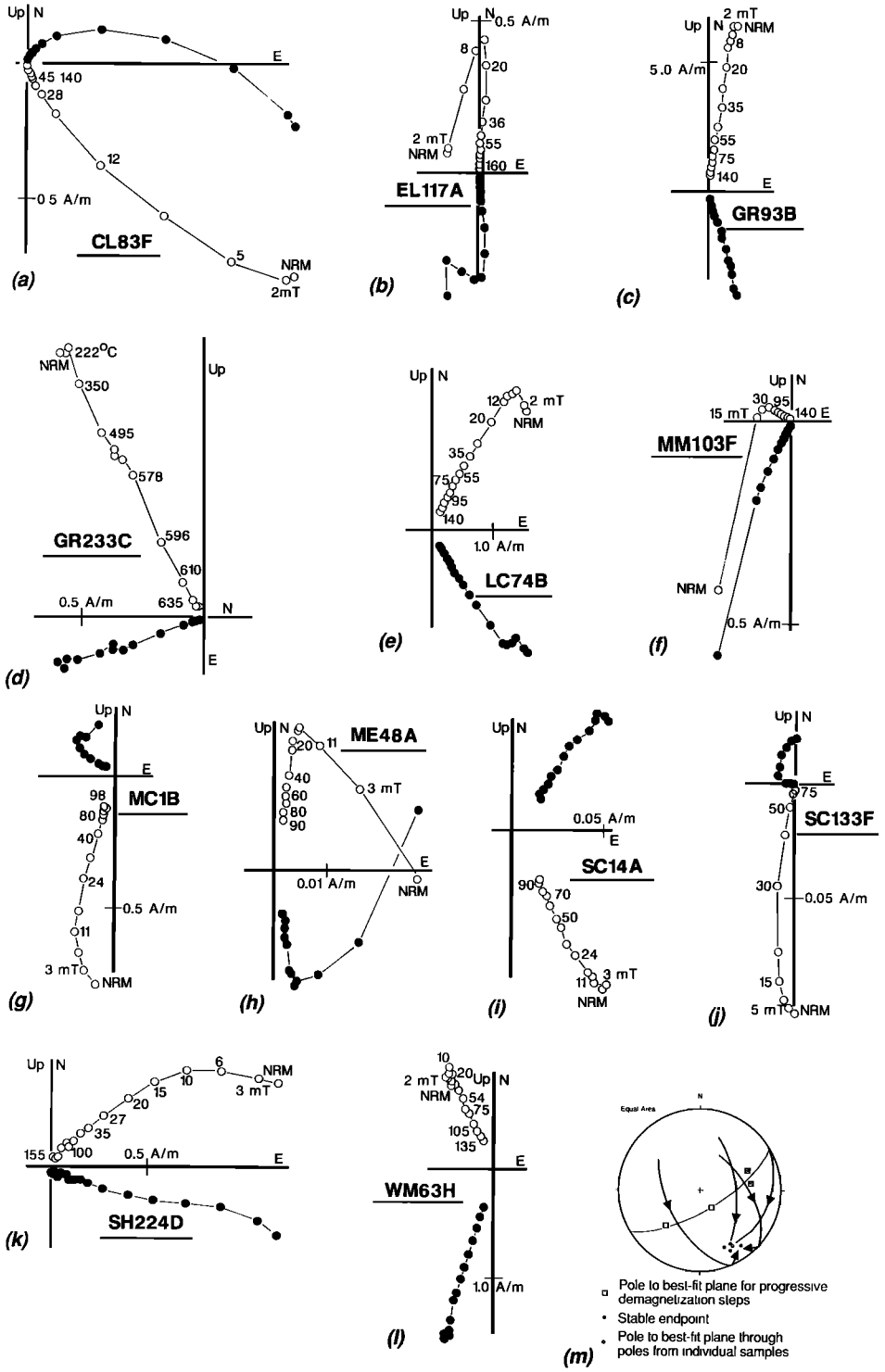
RESULTS

Below we summarize paleomagnetic results presented in Appendix 1 (Tables 1 and 2; Figures 4 and 5). The angular standard deviation of site mean directions and the presence of dual polarity sites indicate that secular variation was clearly averaged at the Sawmill Canyon and Spring Hill localities. Secular variation was less well averaged at the Methodist Creek, Ellis , Massacre Mountain, Copper Lake, Meadow Creek, and Garrett Rock localities (in that order). The Warren Mountain results are equivocal. Secular variation was not averaged at the Long Creek locality (k=83.5).

Directions from five localities (Sawmill Canyon (SC), Methodist Creek (ME), Ellis (E), Copper Lake (CL), and Massacre Mountain (MM)) are statistically indistinguishable from the middle Eocene reference direction. The Sawmill Canyon data are the most compelling, because of the large number of sites (22) and positive reversals test. The best estimate of the precision parameter (k) from the other four localities indicates that these data are also reliable. These concordant directions imply that vertical axis rotation did not accompany tilting and normal faulting in the Lemhi or central Lost River ranges.

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Of the remaining localities two have concordant inclinations and discordant declinations (Warren Mountain and Spring Hill). The dispersion of site means directions and the presence of antipodal dual polarities suggest that the 23 sites at the Spring Hill locality provided a sufficient sampling of paleosecular variation and that undetected initial dip in the volcanic rocks was not a problem at this locality (Table 2, Appendices 1 and 2). The declination at the Spring Hill locality is 43.0° counterclockwise from the expected direction, indicating considerable counterclockwise tectonic rotation.



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Apparent clockwise rotation of 19.8°±16.5° at the Warren Mountain locality conflicts with more reliable results from the nearby Methodist Creek locality. The equivocal tilt correction at the Warren Mountain locality (Appendix 1) prompted us to discount the paleomagnetic evidence for clockwise rotation at this locality.

Three localities exhibit discordant inclinations and declinations (Meadow Creek, Garrett Rock, and Long Creek, See Table 2). Flattening of inclinations at Long Creek is clearly due to insufficient sampling of paleosecular variation (Table 2; Appendix 1). Discordant inclinations at the Garrett Rock and Meadow Creek localities may reflect either insufficient sampling of paleosecular variation or erroneous structural corrections. Declinations are 29.9°±11.3° counterclockwise from the expected direction at the Meadow Creek locality. Data from the Meadow Creek and Spring Hill localities show counterclockwise discordant declinations of 30°-40°.

Interpretation of discordant declinations at the Garrett Rock and Long Creek localities is not straightforward because inclinations are discordant (Table 2; Appendices 1 and 2). Several lines of evidence show that the same lava flows were sampled at both localities (Appendix 1), so we compared the structurally corrected directions at the Long Creek locality with the direction for the Garrett rock locality in addition to comparing each with the time-averaged reference direction. The structurally corrected locality mean declination for Long Creek is rotated approximately 53° counterclockwise relative to the mean for Garrett Rock. This rotation is roughly the amount indicated by the geological relationships alone (Figure 6; Appendix 1). Even though paleosecular variation was probably not fully averaged at the Long Creek and Garrett Rock localities the correspondence between paleomagnetic declinations and geologic observations suggests that these paleomagnetic data still yield information about large yet localized rotations in the central Lost River Range.

Our analysis of possible sources of error and additional uncertainty in the paleomagnetic data (Appendix 2) shows that whereas small rotations may be difficult to evaluate with this data set, rotations greater than ~10°-15° are detectable. Substantial initial tilt is not a problem at any of the 10 localities (Appendix 2).

Regions affected by vertical axis rotations vary in scale. The amount and sense of rotation change radically from one small tilt domain to another, as indicated by the results from Garrett Rock and Long Creek. Consistent patterns of rotation (or lack thereof) across much larger domains are also observed. The entire N-S striking domain of volcanic rocks from Methodist Creek to Sawmill Canyon is unrotated. This domain is 50 to 80 km in N-S extent if the Ellis locality is included. Results from Spring Hill and Meadow Creek define a large domain of counterclockwise rotation in the northern Lost River Range with a dimension along strike of at least 25 km. The absence of significant cross structures in the northern Lost River Range [Mapel et al., 1965] suggests that this domain rotated essentially as a rigid block.

A graphic comparison between the strike of bedding and paleomagnetically determined rotations shows that strike is a good indicator of vertical axis rotation (Figure 7). The Long Creek locality mean is shown with an open symbol because paleosecular variation was clearly not averaged at the locality. Results are somewhat scattered, but in general localities with paleomagnetic evidence for counterclockwise rotation have strikes NNW to NW while the single locality with convincing paleomagnetic evidence for clockwise rotation (Garrett Rock) exhibits a NE strike. Most localities plot near the origin in Figure 6, indicating that rocks in most localities are unrotated and strike north.

Vertical axis rotation is either synchronous with or younger than tilting. Figure 7 suggests that N-S striking Cenozoic rocks throughout east central Idaho are probably not rotated. The NE flank of the Lemhi Range, with its N-S striking Challis strata, may represent a northward continuation of the unrotated domain in the central Lost River and Lemhi ranges (Figure 1). Wells and Hillhouse [1989] previously documented synchron ous rotation and tilting based on their study of extended regions west of the Colorado Plateau.

EVALUATION OF MCKENZIE AND JACKSON BLOCK ROTATION MODEL

Derivation of the Equation

The McKenzie and Jackson [1983, 1986] block rotation model may be used to predict the amount and sense of vertical axis rotation for comparison with the results of our paleomagnetic investigations. A simple geometrical analysis of the McKenzie and Jackson block rotation model shows that the amount of extension across the range-bounding faults depends on (1) the amount of rotation of the blocks, R, (2) the width of the rotating block, W, and (3) the orientation of the block, θ (Figure 3). The equations can be cast either in terms of the initial angle θ_1 between the zone boundary and the fault blocks or the final angle θ_2. We prefer to use θ_2 because this quantity is measurable if the orientation of the zone boundary is known. We assume that two corners of each rotating block are fixed at the zone boundary to derive the the equation relating extension E across block-bounding faults and rotation R.

$$E = W \sin R \{1 / \tan(\theta_2 - R) - \tan(R/2)\} \quad (1)$$

One variation of this block rotation model assumes that crustal blocks are not fixed to the boundaries of the zone but instead float freely within the deforming zone. The rotation rate is halved in this case [McKenzie and Jackson, 1986], but the

Fig. 4. (a-l) Examples of the wide range of response to progressive alternating field and thermal demagnetization by Challis volcanic rocks in the form of representative orthogonal demagnetization diagrams [Zijderveld, 1967] showing the endpoint of the magnetization vector simultaneously plotted onto the horizontal (E-W, N-S) plane (solid circles) and the vertical (E-W or N-S, up-down) plane (open circles) for each demagnetization step. One or two demagnetization diagrams from each locality are shown. Peak demagnetizing inductions (in milliTesla, mT) or degrees Celsius (°C) are given beside projections onto the vertical plane. The intensity of the magnetization, in amperes/meter, is given along one of the axes. (b) Tuff of Ellis Creek, (j) probable tuff of Challis Creek. (m) Equal area projection of the remanent magnetization directions recorded during progressive thermal demagnetization of specimens from site GR230, which appears to have been struck by lightning. Analysis of demagnetization data from this site was completed using both stable endpoint and remagnetization circle data.

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Table 2a. Locality Mean Directions and Statistics

Locality	In Situ Dec	In Situ Inc	Corrected Dec	Corrected Inc	α_{95}	k	R	N/N_o	R±ΔR	F±ΔF	Strike	Dip
Copper Lake	304.2	37.1	336.0	57.2	10.7	17.5	11.3724	12/12	-14.0±16.2	6.6±8.7	354.0	34.0 *
Ellis Creek	168.4	-61.1	159.8	-49.0	10.2	35.7	6.8319	7/7	-1.6±17.1†	3.2±8.2†	227.7	13.5 *
Garrett Rock §	161.1	-22.3	184.9	-52.0	7.7	30.0	12.6035	13/13	**14.9±10.5**	**11.9±6.4**	36.5	41.3 *
Long Creek §	117.2	-5.2	132.3	-47.5	4.8	83.5	11.8680	12/12	**-37.7±6.6**	**16.3±4.2**	0.0	50.0
Massacre Mountain	313.5	35.8	359.2	55.5	10.4	18.2	11.3964	12/14	9.2±15.1	8.3±8.4	0.0	45.0
Meadow Creek §	118.5	-51.4	140.1	-54.2	7.9	34.8	10.7129	11/12	**-29.9±11.3**	**9.9±6.5**	320.2	16.6 *
Methodist Creek	131.4	-39.5	177.2	-58.2	11.0	26.3	7.7342	8/9	7.2±16.9	5.5±8.8	358.6	40.9 *
Sawmill Canyon	297.8	34.0	350.2	65.4	7.1	20.0	20.9682	22/22	0.2±14.2	-1.5±6.0	358.0	48.6 *
Spring Hill §	100.6	-34.0	127.0	-61.6	6.2	25.2	22.1273	23/26	**-43.0±11.1**	2.3±5.3	342.8	35.1 *
Warren Mountain §	148.9	-53.0	189.8	-55.5	11.4	21.3	8.6253	9/9	**19.8±16.5**	8.2±9.2	354.0	28.3 *
All			341.6	57.5	8.2	39.8	9.7490	10/10	-8.4±12.6	6.5±6.8		
Selected localities			348.1	57.4	7.8	122.8	4.9595	5/10	-1.9±11.7	6.6±6.3		

Dec and Inc are declination and inclination of locality mean in degrees; α_{95} is radius of 95% confidence circle in degrees; k is estimate of Fischer precision parameter for locality mean magnetization parameter; R is length of resultant vector; and N/N_o is number of site means included in statistical analysis/number of site means analyzed. R±ΔR is vertical axis rotation in degrees and associated 95% confidence limits; F±ΔF is flattening, in degrees, and associated 95% confidence limits using the methods of Beck [1980] and Demarest [1983]. Strike and dip indicate structural correction (right-hand rule). 350°, 63.8°, A_{95}=3.0° reference direction from Diehl et al., 1983. Bolded entries are statistically significant.

* Mean of structural data.

† Based on in situ direction. See discussion in text. Tilt-corrected R±ΔR=10.2±12.7 and F±ΔF=15.3±8.2.

§ Locality excluded from selected group.

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Table 2b. Locality Mean Statistics

	CSD	α_{13}	α_{12}	α_{12}/α_{13}	lat	long
Copper Lake	25.9	6.7	16.7	2.24	44.10	113.58
Ellis Creek	16.4	6.0	12.9	2.15	44.68	113.99
Garrett Rock	17.8	4.3	10.8	1.93	44.23	113.59
Long Creek	9.7	3.0	5.8	1.93	44.17	113.58
Massacre Mountain	23.4	5.3	15.4	2.90	44.13	113.50
Meadow Creek	18.4	2.8	12.1	4.30	44.46	113.97
Methodist Creek	19.3	6.0	14.5	2.40	43.98	113.47
Sawmill Canyon	26.3	8.6	10.6	1.23	44.31	113.38
Spring Hill	20.8	6.7	8.3	1.24	44.27	113.75
Warren Mountain	22.7	8.0	15.6	1.95	44.00	113.50

CSD is estimated circular standard deviation of virtual geomagnetic poles (VGPs). α_{13} and α_{12} are approximate 95% confidence angles using Bingham statistics [Onstott, 1980] for VGPs; lat and long are latitude and longitude of locality in degrees.

equation relating rotation and extension still holds.

The spacing between range-bounding faults, W, is about 30 km in this area, and the ranges are nearly perpendicular to the trend of the Snake River Plain (θ_2 is about 85°-90°). Figure 3b shows that extension varies in a nonlinear way with the amount of vertical axis rotation. One kilometer of extension corresponds to about 10°-15° of vertical axis rotation. Five kilometers of horizontal extension, however, corresponds to more than 27°-33° of clockwise rotation. Amounts of vertical axis rotation are computed directly from estimates of extension (heave) across the active Basin and Range faults using equation (1), assuming that block rotation began at the inception of the Basin and Range faulting in the Lost River and Lemhi ranges.

Slip Estimates Across the Lost River Fault

Estimates of extension across Basin and Range faults are difficult to constrain because the hanging wall is typically buried beneath Cenozoic basin fill. Table 3 presents the best estimates of the dip slip and total heave across the Lost River fault compiled or modified from various sources (locations A, D, E, F2, G1, G2), together with new estimates (locations B, C, F1, G3) based on offset Paleozoic and Cenozoic strata and consideration of the available gravity data [Webring and Mabey, 1987]. Unpublished cross sections constrain estimates of heave at locations B and C.

In the Borah Peak area (Figure 1, Table 3, location A) a series of NW striking normal faults within the footwall of the Lost River Fault have nearly 4 km of dip separation across them. Some of these faults were activated during the 1983 Borah Peak earthquake and are clearly part of the range front fault system [Susong et al., 1990]. An additional unknown amount of slip must be localized along the range-bounding fault. Gravity data show that the adjacent basin fill is relatively thick [Mabey, 1985; Wilson et al., 1990] so an additional 0.9-1.3 km of throw is inferred. The heave across the fault is estimated to lie between 2.4 and 3.2 km.

At location B the Lost River fault splays into three branches at the Mackay-Pass Creek segment boundary (Figure 1, Table 3, location B). By matching formations in the hanging wall and footwall of these splays and using gravity data [Crosthwaite et al., 1970; Bankey and Kleinkopf, 1988] to estimate the thickness of the basin fill, we calculate 3.5-3.9 km of heave across the three strands of the Lost River fault at this location.

A cross section constructed a few kilometers northwest of location B at the Mackay Narrows suggests a minimum of 2.4-2.9 km of heave across the Lost River fault (location C). Exposures of upper Paleozoic bedrock in the hanging wall of the Lost River fault project up through a very thin alluvial cover in this area, so slip estimates are unusually well constrained.

Two estimates of slip along the southern end of the Lost River fault yield fault heaves between 3 and 4 km (Figure 1, Table 3, locations F1 and F2). Both slip estimates are based on offsets of the Tertiary/Paleozoic unconformity. Gravity data from Crosthwaite et al. [1970] constrain the hanging wall structure in cross section F1.

Numerous workers have estimated the minimum throw across the Thousand Springs segment of the Lost River fault (Figure 1, location G) using a variety of geological , topographic, and geophysical data. Estimates of heave range from 2.1 to 2.9 km (Table 3).

Taken as a whole these data suggest minimum heaves across the Lost River fault ranging from about 2 to 4 km. The 2-4 km of fault heave along the Lost River fault corresponds to about 15°-30° of clockwise rotation in the McKenzie and Jackson model (Figure 3). Adjusting the angle θ_2 and block width (W) will alter this prediction somewhat, but in any case a significant measurable amount of vertical axis rotation is predicted. Rotations of 15°-30° are well within the detection limits of paleomagnetic investigations (Appendix 2). We did not estimate the heaves across the Lemhi and Beaverhead faults, but gravity data suggest these are likely to be greater than the heave across the Lost River fault [Webring and Mabey, 1987; Bankey et al., 1985; Bankey and Kleinkopf, 1988]. Thus the predicted uniform magnitude vertical axis rotation in east central Idaho may be greater than 15°-30° .

Comparison With Model Predictions

Paleomagnetic results from 10 localities in the Lost River and Lemhi ranges (Table 2) do not support McKenzie and Jackson's [1986] prediction of uniform sense and magnitude clockwise rotation in east central Idaho. Anders et al. [1988] also evidently found no evidence for vertical axis rotation in their preliminary study of Miocene and younger basalts and ash flow tuffs east of the Lemhi Range. Left-normal oblique-slip along the Thousand Springs segment of the Lost River fault, and other segments of similar trend or slip-sense in the region,

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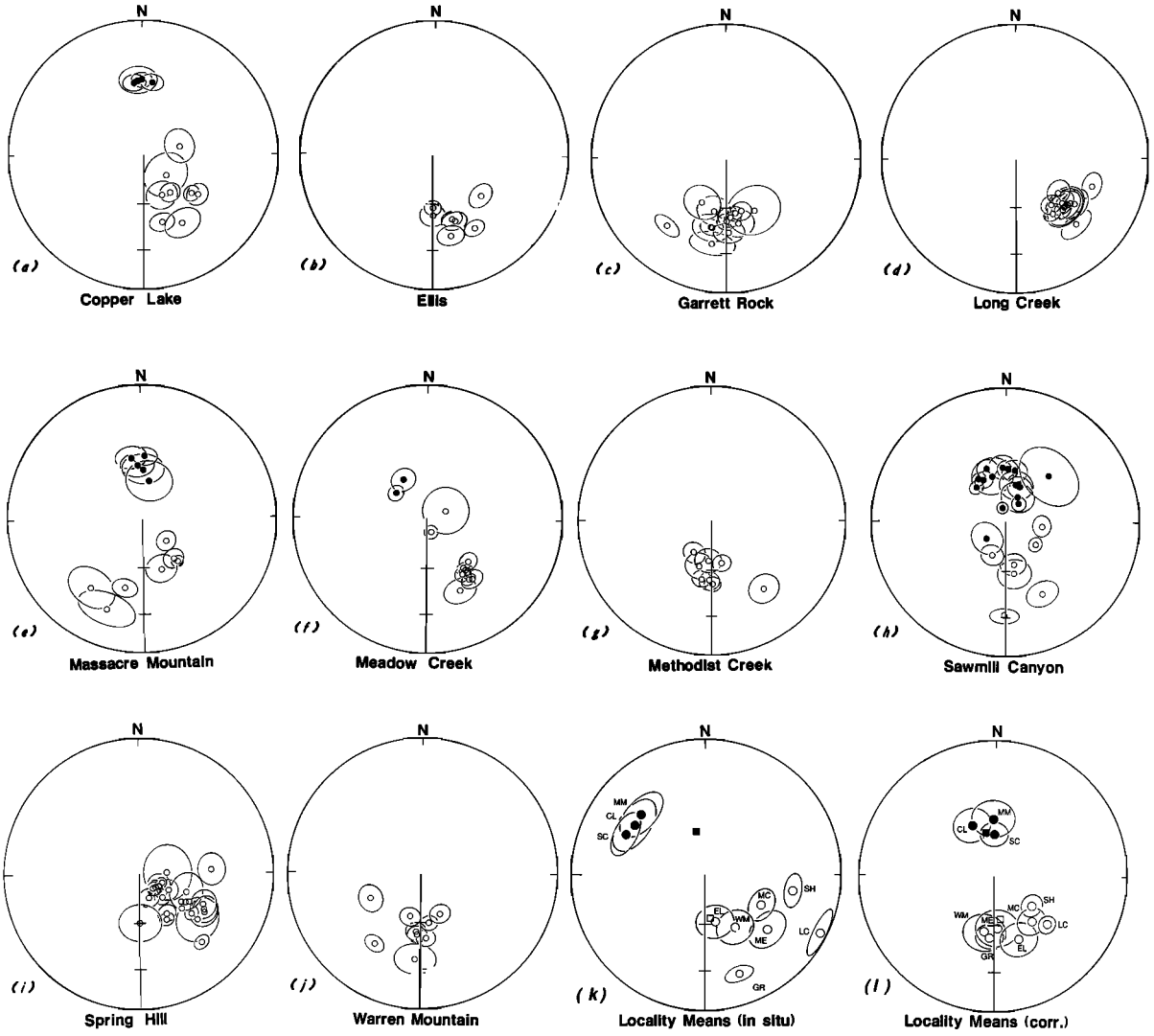


Fig. 5. (a-j) Equal area projections of structurally corrected site mean directions for each locality, with projected cones of 95% confidence. In these and all other equal area projections, open (solid) symbols refer to the upper (lower) hemisphere. Structural corrections are listed in Table 2. Mean directions from sites which have been excluded from statistical analysis at the locality level are not shown in these projections (except site MC12). Expected direction shown in 5k and 5l. (k and l) Equal area projections of the 10 locality mean directions, in situ and structurally corrected coordinates, respectively. Expected direction denoted by square. Note that concordant in situ direction from the Ellis locality becomes discordant after tilt correction. Locality names as in Figure 1. Data are reported in Table 2.

therefore, does not necessitate clockwise block rotation. Left-normal slip along the Thousand Springs segment probably does not characterize the length of the fault (as implied by McKenzie and Jackson [1986]) and may develop locally because fault segments are not orthogonal to the regional N45°E trending minimum principal stress [Stickney and Bartholomew, 1987]. We infer that a left-lateral component of slip characterizes only those segments of the Lost River, Lemhi, and Beaverhead faults which strike northerly. Pure normal slip occurs along segments perpendicular to the least compressive stress, and components of right-slip characterize segments striking easterly. We know of no geologic and geophysical observations that either support or contradict this interpretation. The interpretation is testable, however, by kinematic analysis of the active fault segments.

More recently, Westaway [1989] extended and elaborated on the McKenzie and Jackson model. Westaway [1989] proposed that regions adjacent to the eastern Snake River Plain, such as east central Idaho, may have experienced one sense of vertical axis rotation followed by the opposite sense of rotation as

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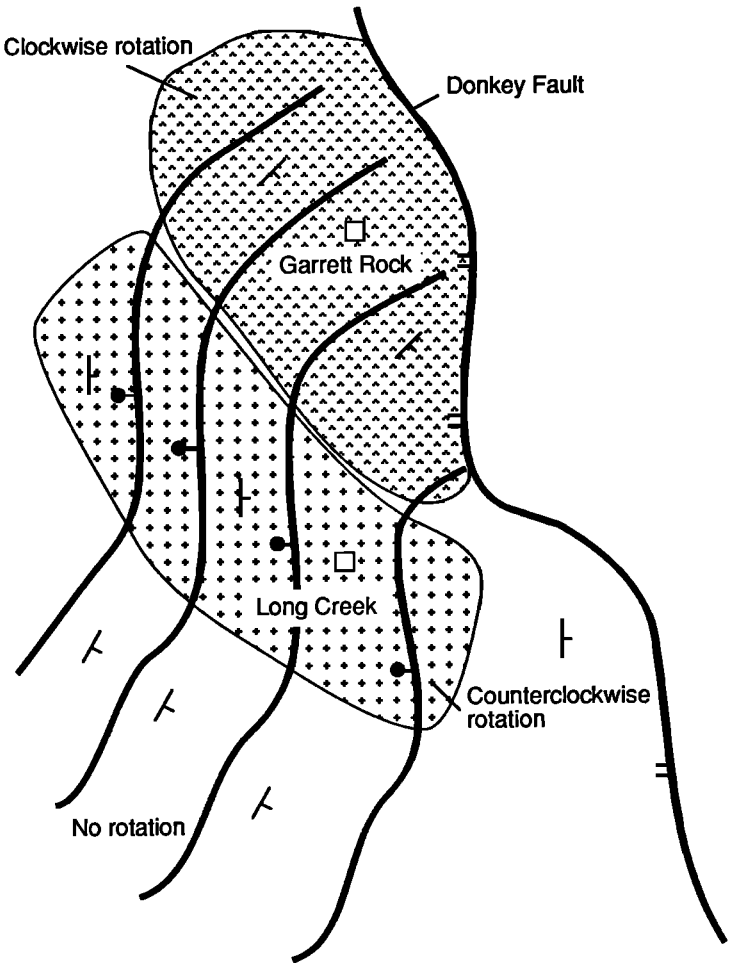


Fig. 6. Schematic geologic map showing relationships at Garrett Rock and Long Creek localities. Bold lines denote normal faults. Shaded regions show areas affected by vertical axis rotation. A southeast plunging fold may separate the two domains.

North America passed southwestward over the Yellowstone hotspot in late Cenozoic time. Depending on the current position of an area relative to the hotspot, the model predicts net clockwise, net counterclockwise rotation, or under some conditions, no net rotation [Westaway, 1989]. This model is difficult to test paleomagnetically because it does not predict the amount and sense of net rotation at any location in space and time [Westaway, 1989].

MECHANISMS FOR DIFFERENTIAL TECTONIC ROTATION

The large domain of counterclockwise rotation in the northern Lost River Range is not predicted by existing kinematic models of the area. Possible mechanisms for differential tectonic rotations between the northern and southern areas of the Lost River Range include drag folding, small independently rotating blocks, a modified McKenzie and Jackson block rotation model, and rotation of the hanging wall of a low-angle normal fault. We are not attempting to model rotations of localized blocks at Garrett Rock or Long Creek; rather we are interested primarily in explaining the rotations (or lack thereof) and spatial patterns of large crustal blocks.

Drag: Drag folding or block rotations adjacent to the Trans-Challis fault zone [e.g,. Ekren, 1985; Kiilsgaard et al., 1986] may explain the observed counterclockwise rotation of localities in the northern Lost River Range (Figure 8a). McIntyre et al. [1982], and Ekren [1988], presented evidence for left lateral offset along NE striking faults in the Trans-Challis fault zone ~20 km NW of the Ellis location. If significant left-lateral slip was accommodated along this zone, then counterclockwise rotation by drag folding or block rotation is possible in adjacent areas. This model predicts that rotation decreases smoothly southeastward from the Trans-Challis fault zone and is not supported by the paleomagnetic data. The Ellis locality in the north Lemhi Range is closest to the Trans-Challis fault zone yet shows no apparent vertical axis rotation. The Meadow Creek locality, to the south, exhibits approximately 30° of counterclockwise rotation. In order to explain the counterclockwise rotation at Spring Hill,

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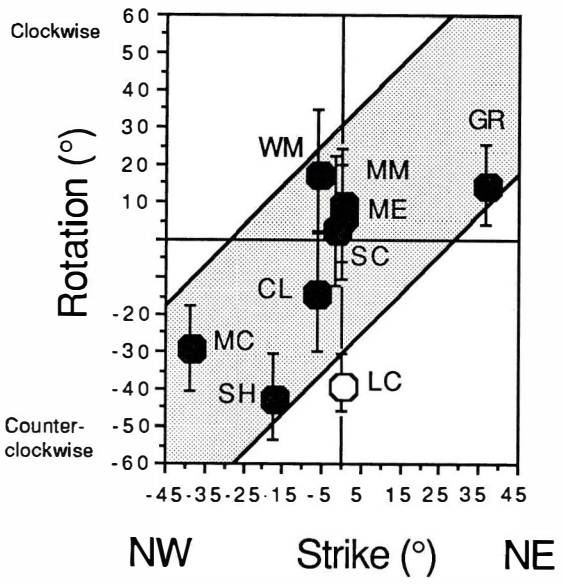


Fig. 7. Tectonic rotation versus strike for each locality except the flat-lying Ellis locality. Locality name abbreviations as in Figure 1. Vertical bars represent the 95% confidence intervals (ΔR) on the discordant declination given in Table 2. Long Creek locality is shown as an open circle because it is very unlikely that secular variation was averaged at this locality. In general, the strike of bedding is a good indicator of the sense of vertical axis rotation at each locality. Strikes east of north correspond with clockwise rotation, whereas strikes west of north correspond to counterclockwise rotation. The localities which cluster in the center of the graph (with northerly strikes) experienced little vertical axis rotation.

drag would have to extend more than 60 km from the southernmost strand of the Trans-Challis fault zone.

Transtension: A modified form of the McKenzie and Jackson model could explain the observed heterogeneous rotations (Figure 8b). The spatial distribution of rotated and unrotated localities could be explained if the Meadow Creek and Spring Hill localities resided on counterclockwise rotating blocks bounded by normal oblique faults while localities to the SE, east, and north were anchored to country rock outside the shear zone and remained unrotated. Rotated and unrotated localities are best accommodated by a northeasterly striking shear zone extending from the northern Lost River Range into the northern Lemhi Range. In this model, 30°-40° of counterclockwise rotation is predicted in the northern Lemhi Range between the Ellis and Sawmill Canyon localities. This mechanism cannot explain the consistent north strike of volcanic rocks all along the NE flank of the Lemhi Range [Ruppel, 1968, 1980] from 44° 30' N to 44° 45' N (Figure 1).

Fragmentation into small blocks: Nelson and Jones [1987] proposed that the crust NE of the Las Vegas Valley shear zone is broken into small irregularly shaped blocks each of which rotated different amounts in response to mantle vorticity (their Figure 6e). The depth extent of blocks in the study area in Idaho is probably limited by one or more basal detachment faults, so the small block model of Nelson and Jones [1987] would have to be modified to apply to east central Idaho (Figure 8c). One might expect greater magnitudes of rotations in areas where crust is broken into smaller, more equant blocks [Lamb, 1987]. This prediction may be appropriate for our results at Garrett Rock and Long Creek but can not be used to explain data from the northern Lost River Range where consistent rotations affected a large area. The Massacre Mountain and Copper Lake localities are also located in small blocks yet are unrotated. Overall the correlation between size of fault-bounded block and amount of vertical axis rotation is poor.

Table 3. Slip on the Lost River Fault

Locality	Dip Separation, km	Throw, km	Heave, km	Source
Borah Peak horst	3.7-4.9	2.8-3.7	2.4-3.2	A
Swauger Gulch	5.5-6.0	4.2-4.6	3.5-3.9	B
Mackay Narrows	4.5-4.9	3.5-3.8	2.4-2.9	C
Willow Creek Hills	2.75-3.1	2.1-2.4	1.8-2.0	D
Warm Springs segment	≥3.2	≥2.5	≥2.1	E
Pass Creek segment, south end	4.5-5.7	3.5-4.8	2.9-3.7	F1
Arco segment	5.0-6.0	3.8-4.6	3.2-3.9	F2
Thousand Springs segment		2.5-2.8	2.1-2.3	G1
Thousand Springs segment		3.0	2.5	G2
Thousand Springs segment		3.5	2.9	G3

Fault dip of 50° assumed in all calculations of throw and heave. Source of estimate or source of data: A, Susong et al. [1990] cross section A-A'; Gravity data from Mabey [1985] and Wilson et al. [1990]; B, Data from S.U. Janecke, unpublished mapping, 1987-1989, Crosthwaite et al. [1970], Bankey et al. [1985], and Bankey and Kleinkopf [1988]; C, Data from S.U. Janecke, unpublished mapping, 1987-1989, and Nelson and Ross [1969]; D, Data from Mapel et al. [1965] and Skipp and Harding [1985]; E, Cross section B-B' of Mapel et al. [1965] modified to include a 50° dipping fault; F1, Data from Rember and Bennett [1979b] and Crosthwaite et al. [1970]; F2, Skipp and Hait [1977], cross section A-A' in Figure 1; G1, Crone and Machette [1984, p. 664]; G2, Skipp and Harding [1985, p. 699]. ~3.0 km of Late Cenozoic offset, (≥ 5.0 km total); G3, Maximum relief of Lost River Range plus ~1.5 km valley fill in Thousand Springs Valley [Mabey, 1985, p. 684].

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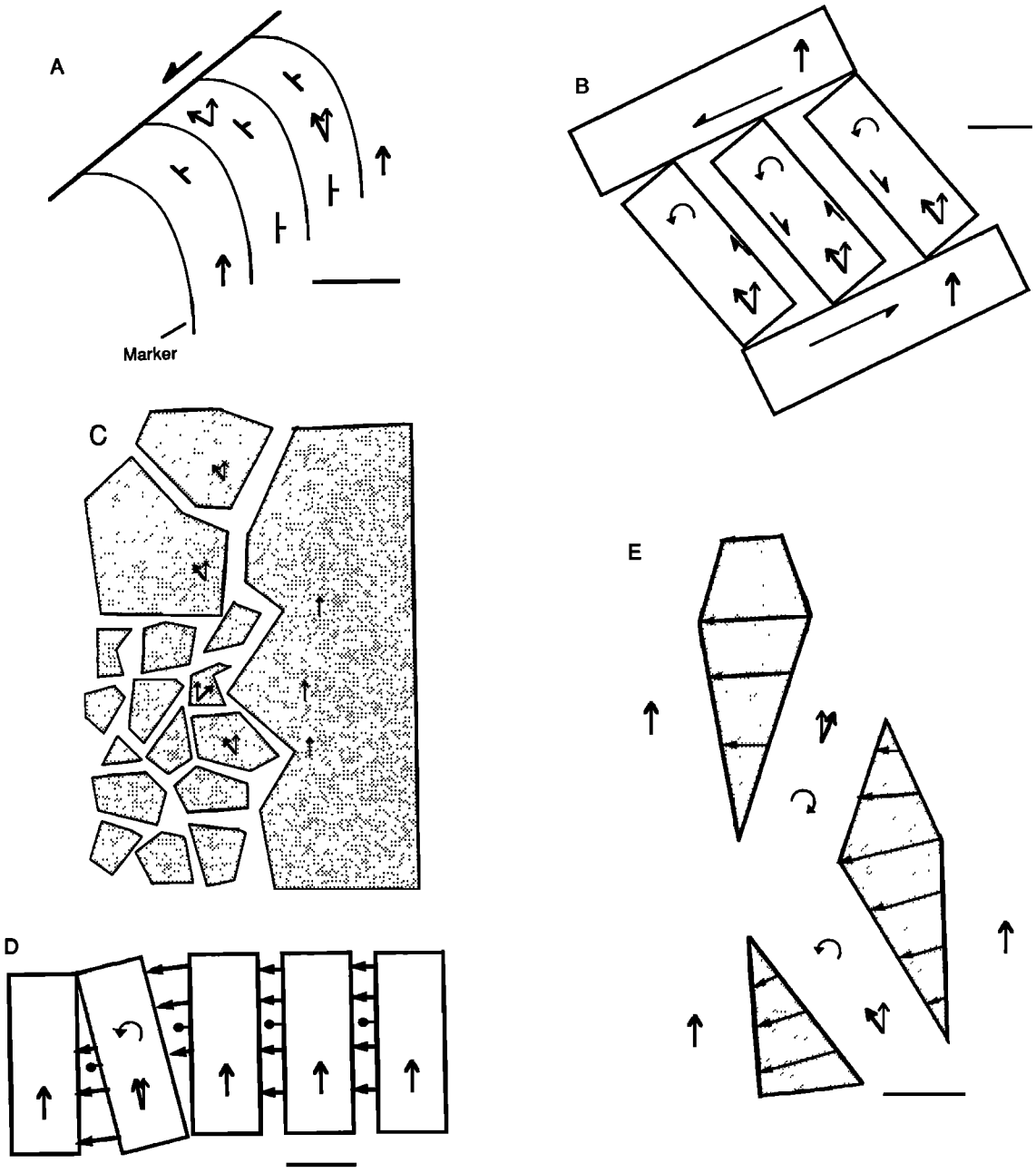


Fig. 8. Possible mechanisms to explain heterogeneous vertical axis rotation. (a) Drag adjacent to a strike-slip fault. (b) Left-lateral McKenzie and Jackson [1983, 1986] transtension model. (c) Small block model modified from Nelson and Jones [1987]. (d) Hanging wall rotation model. (e) Variation of hanging wall rotation model which predicted both clockwise and counterclockwise rotation associated with diamond-shaped (in plan view) basins. In each diagram the bold open-ended arrows denote the magnetization direction, the lighter open-ended arrows indicate the expected direction of magnetization. For convenience the expected direction is shown directed due north. Closed arrows show the horizontal projection of slip vectors on faults. Scale bar denotes tens of kilometers.

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Hanging wall rotation: Differential rotation of the hanging wall of the Donkey low-angle normal fault relative to its footwall is our preferred model for explaining the paleomagnetic data and is supported by geologic observations (Figure 8d, Figure 9). Rotation is predicted within the Lost River Range north of Copper Lake and in areas farther to the west, extending at least to the next N-S normal fault in the system (Figure 8d). This simple kinematic model explains the NNW strike of the flows at the Spring Hill and Meadow Creek localities and explains the absence of rotation in the Lemhi Range and most of the central Lost River Range. One appealing aspect of this model is that when the northern Lost River Range is restored to a prerotation configuration, flows along the entire length of the Lost River Range dip uniformly toward the east, except in a few fairly localized and structurally complicated areas in the central Lost River Range (Figure 10).

Differential rotation across the Donkey fault opened a triangular north plunging basin between the present Lost River and Lemhi ranges. Slip across the fault is difficult to constrain but must exceed 13 km in the central Donkey Hills based on offset Paleozoic strata. Slip is negligible at the southern termination of the Donkey fault. East dipping Tertiary strata in the footwall and hanging wall of the fault suggests that the fault itself has been tilted about subhorizontal axes to its present low angle. Dates from one tuff interbedded within syntectonic conglomerate sequences shed from the footwall of the fault suggests slip on the Donkey fault was ongoing at about 30 Ma in Oligocene time [Janecke and Snee, 1990]. The minimum age of faulting is not known, but regional relationships indicate slip ended prior to the onset of Basin and Range faulting at about 16-17 Ma [Fields et al., 1985; Burbank and Barnosky, 1990].

Some support for the hanging wall rotation model comes from the orientation of Mesozoic to early Tertiary (pre-Challis) fold axes mapped in the study area. Trends of fold axes in part support postfolding counterclockwise rotation of the northern and central Lost River Range relative to adjacent areas. The areas of northwest fold axes do not coincide completely with areas which exhibit counterclockwise rotation, however, and the amount of rotation predicted by our analysis of fold orientations is less than that detected paleomagnetically. The counterclockwise rotation implied by the orientation of the fold axes, assuming that axes were initially subparallel throughout the region, is less than 30°.

Localized shortening is predicted near the southern end of the tapering Donkey fault to accommodate interaction between the rotating hanging wall block and the fixed unfaulted region to the south. The strikes of volcanic strata at the Garrett Rock, Long Creek, and Copper Lake localities crudely define SE plunging folds of postmiddle Eocene age (Figure 7). These

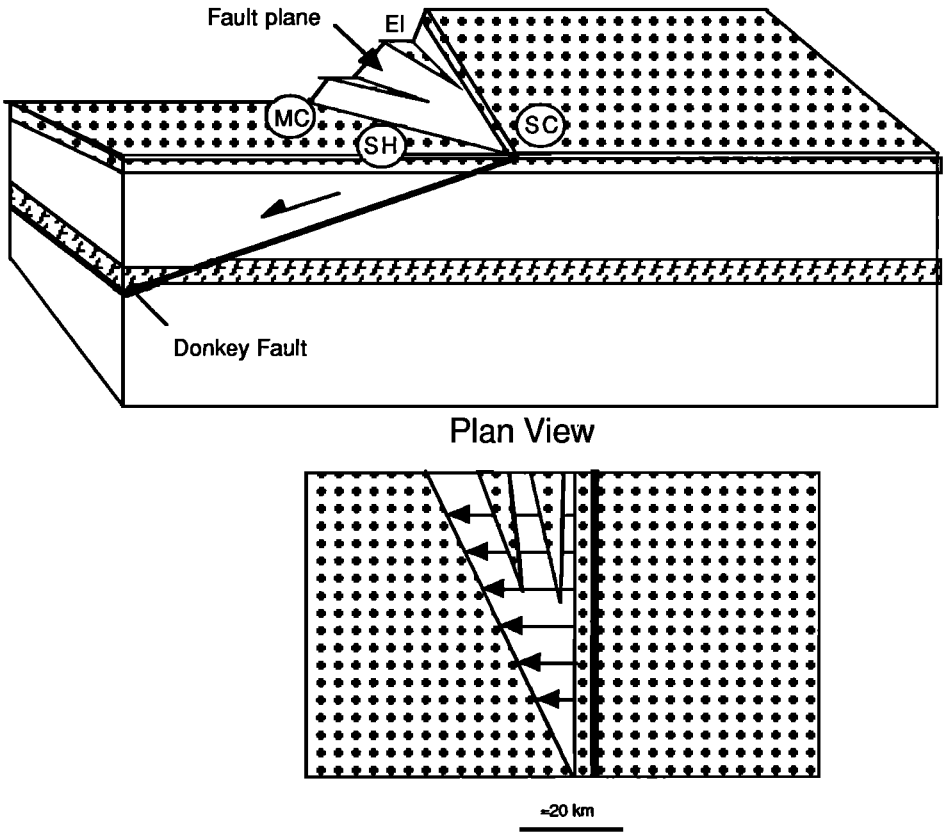


Fig. 9. Block model showing three-dimensional geometry of hanging wall rotation model in Figure 8d. The Donkey fault tapers to the south. Location of selected localities is shown schematically. Very approximate scale is indicated.

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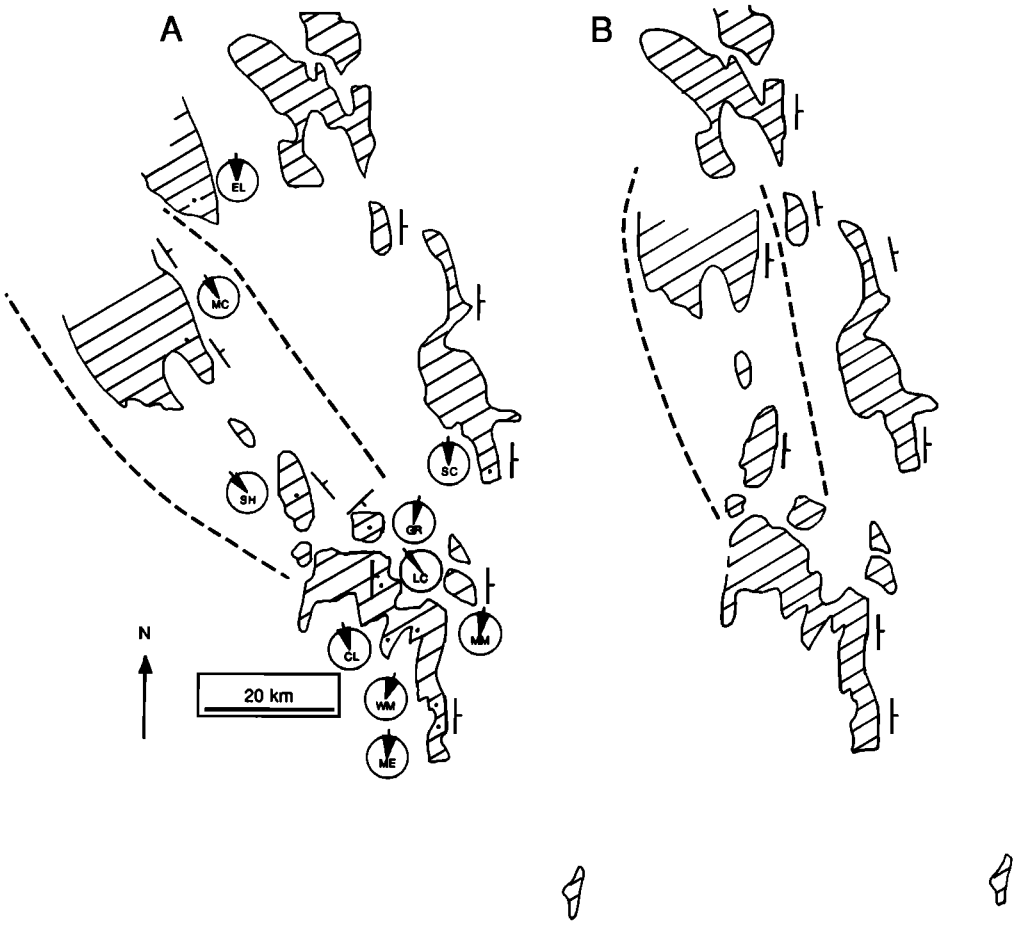


Fig. 10. (a) Simplified geologic map showing the present distribution of volcanic rocks in the Lost River Range and Lemhi Range (diagonal rule), approximate strikes and dip directions of Tertiary strata, and tectonic rotations with their associated uncertainties. Locality name abbreviations as in Figure 1. Dashed lines enclose rotated domain in northern Lost River Range. (b) Reconstructed distribution of volcanic rocks assuming 30° of counterclockwise rotation of the northern Lost River Range. Roughly N-S strikes characterize the entire reconstructed region. Relations in the northern part of the area are uncertain.

folds may record NE-SW contraction adjacent to the Donkey fault near its southern end, in a transition zone between rotated and unrotated domains. Thus the hanging wall rotation model may explain the anomalous Garrett Rock and Long Creek results as well as the rotation at the Meadow Creek and Spring Hill localities.

The hanging wall rotation model predicts tens of kilometers of extension at the northern end of the Pahsimeroi Valley and regions to the north (Figure 1), if displacement increases northward as depicted in Figures 8d and 13. The available bedrock mapping and gravity data suggest only moderate amounts of roughly E-W extension in this area [McIntyre and Hobbs, 1987; Fisher et al., 1983; Webring and Mabey, 1987], suggesting perhaps that displacement may taper (i.e. decrease) to the north (shown schematically in Figure 8e).

The model makes numerous testable predictions about Cenozoic deformation in east central Idaho. These include the absence of rotation in the northeastern Lemhi Range and consistent counterclockwise rotation west of the Lost River Range extending to the next major N-S striking normal fault. Samples from over 90 flow units at an additional four localities have already been collected to help delineate the lateral extent of the domain of counterclockwise rotation in the northern Lost River Range and to the west.

The apparent absence of rotation in most of the Lost River and Lemhi ranges could be the consequence of two opposite-sense phases of rotation (e.g.) counterclockwise rotation followed by the same amount of clockwise rotation, or vice versa). Challis volcanic rocks are the only Cenozoic rocks exposed in the study area suitable for paleomagnetic study, so comparison of rotations in younger and older Cenozoic sequences is not possible. Preliminary results reported by Anders et al. [1988] from Miocene and younger volcanic rocks east of the Lemhi Range, however, apparently rule out late

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Cenozoic vertical axis rotation, suggesting to us that the absence of rotation over such a large region is probably not the fortuitous result of two phases of equal but opposite rotation.

The hanging wall rotation model is appealing because it integrates paleomagnetic data with a specific large-displacement fault in the area. This mechanism is realistic geologically and mechanically and results in crustal block rotations in extending terrains without sets of strike-slip faults. Thus crustal thinning and block rotations can be integrated into a coherent kinematic model of continental extension. This mechanism may be generally applicable in the Basin and Range province. Bartley and Glazner [1989] independently suggested a similar mechanism to explain clockwise rotations reported in a paleomagnetic study by Ross et al. [1989] in the Mojave Desert region. A similar mechanism may also explain differential rotation between an eastern and western domain reported by Hagstrum and Gans [1989] in east central Nevada and rotations documented by Wells and Hillhouse [1989] in the Colorado River extensional corridor.

It is only in the context of a comprehensive structural and tectonic study that we are beginning to understand the kinematics of the fault network which produced the heterogeneous block rotation revealed by paleomagnetic data. Integrated paleomagnetic and detailed structural studies are necessary in complexly deformed terrains to decipher the three-dimensional kinematics of continental extension.

CONCLUSIONS AND IMPLICATIONS

Our paleomagnetic study shows that vertical axis rotation locally accompanied Eocene to Oligocene extension in east central Idaho. The rotations are heterogeneous in sense and magnitude, with counterclockwise rotations restricted to a domain in the northern Lost River Range. Most of the region did not rotate during extension. Models of extension which predict uniform sense vertical axis rotation of blocks within a given domain [McKenzie and Jackson, 1986; Ron et al., 1984; Garfunkel and Ron, 1985] do not adequately reflect complex strain paths during crustal extension in Idaho.

Strikes of volcanic strata are good indicators of relative rotations from locality to locality and from tilt domain to tilt domain. Our data indicate that the unrotated reference domain is not the northern Lost River Range, where Challis strata are gently tilted, as we had initially suspected. Strikes of blocks must be compared instead with the large east dipping domain of volcanic rocks in the central Lost River and Lemhi ranges. Vertical axis rotation appears to have been synchronous with (or younger than) tilting along Eocene-Oligocene normal faults.

Paleomagnetic documentation of block rotations depends on averaging paleosecular variation and on well-constrained, accurate tilt corrections. We approached success in averaging paleosecular variation when we were able to sample more than 20 flows. Fewer sites at one locality (e.g.), Long Creek) were often not sufficient for adequately sampling paleosecular variation. Because small uncertainties in the dip of the strata (± 5°) at localities can yield apparent rotations in excess of 10°, we are not confident in our ability to discriminate between 0° and 10° of rotation. Larger rotations are readily detected, however, and allow us to discount the McKenzie and Jackson [1986] model of uniform sense and magnitude rotations for east central Idaho and adjacent Montana. We propose instead that differential rotation of the hanging wall of an Eocene-Oligocene west dipping low-angle normal fault relative to its footwall explains the pattern of block rotations in the area. This kinematic model is generally applicable to extending regions and provides a mechanism for combined crustal thinning and variable sense and magnitude block rotation. Rather than demanding the existence of strike-slip faults to explain vertical axis rotations, the model implies that rotations are produced by waxing and waning of slip along large low to moderate angle normal faults. This is in the spirit of the normal fault ideas of McKenzie and Jackson [1986].

APPENDIX 1

Locality by locality discussions of the paleomagnetic data are presented below.

Copper Lake: We sampled 12 lava flows and one tuff at the Copper Lake locality (Figure 5a). The tilt correction is based on measurements of compaction foliations from tuffs in the section, sightings, and three-point solutions. Samples from both normal and reverse polarity sites generally responded well to alternating field demagnetization (Figure 4a). The site mean data fail a reversals test (Figure A1). Declinations of tilt-corrected directions from the four normal polarity sites are clockwise from those of reversed polarity sites in younger lavas (Figure A1, Table A1). The directions of the normal polarity sites are very tightly grouped, however, and may represent only one time-distinct field direction (see discussion of this problem by Calderone et al.[1990]). If so, a comparison of a single "effective" normal polarity direction with several more dispersed reversed polarity site means may not be valid. The precision parameter is 17.5, and neither the rotation ($R\pm\Delta R$=-14.0±16.2°) nor the flattening ($F\pm\Delta F$=6.6±8.7°) is statistically significant at the 95% confidence limit.

Ellis: Seven sites collected in the Tuff of Ellis Creek and overlying lava flows at the Ellis locality were analyzed (Figure 5b). Samples from seven additional sites collected stratigraphically above site El121 displayed such erratic demagnetization behavior that analysis was aborted and the data were omitted from Table 1. All flows were reversely magnetized (e.g., Figure 4b). The low number of sites and the absence of reversals could indicate that secular variation was not averaged at the locality. The precision parameter is not extremely high (k=35.7), however, and the in situ direction does not differ at the 95% confidence limit from the expected middle Eocene direction (Table 2). Tilt correction based on three measurements from a single exposure of volcaniclastic rocks within the section displaces the observed direction, away from the expected direction and results in a significant inclination flattening ($F\pm\Delta F$=15.3°±8.2, Table 2). The entire sequence, however, appears to be essentially flat lying and untilted, and we infer that the attitude of the volcaniclastic rocks is not representative of the locality. Rotation and flattening values in Table 2 are based on the *in* situ remanence direction rather than the structurally corrected direction.

Garrett Rock: We sampled 13 strongly silicified, hematitic, and autobrecciated lava flows at Garrett Rock (Figure 5c). Most samples responded well to alternating field demagnetization, and remanence directions from individual sites are well grouped (Figures 4c and 4d). Characteristic magnetization of each site was apparently acquired after breccia blocks stopped moving relative to one another. Remagnetization circle analysis at two apparently lightning-struck sites successfully isolated a primary remanence direction (Figure 4m). All flows were reversely magnetized and a statistically significant flattening of tilt-corrected inclination

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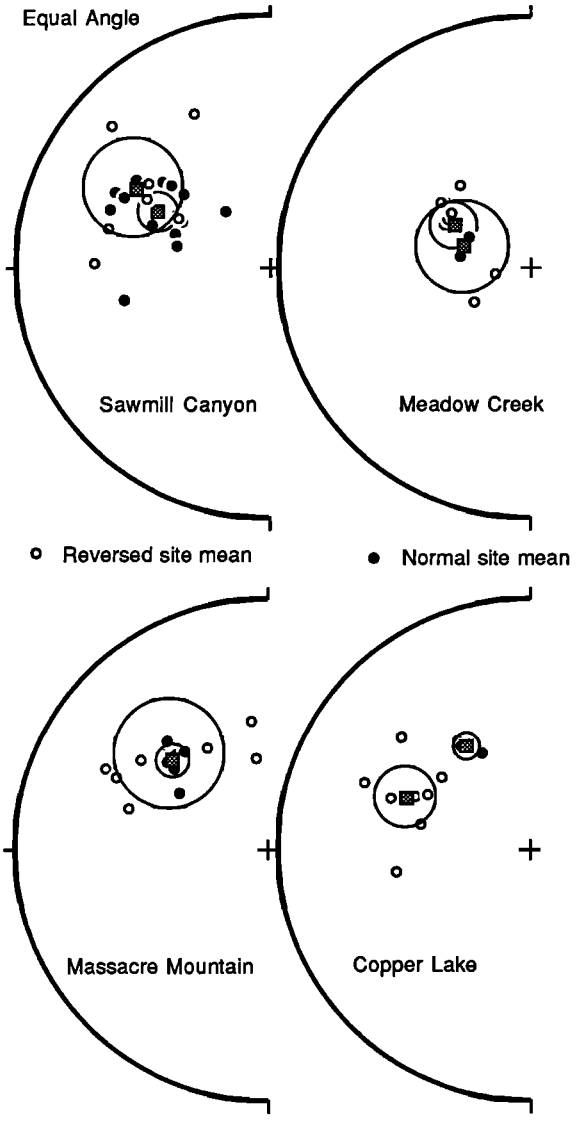


Fig. A1. Lower hemisphere equal angle projections of site mean directions comparing group means of the normal polarity sites with group means of reversed polarity sites. Means (squares) are plotted with their associated α_{95}. Open symbols denote reverse polarity site means, solid symbols denote normal polarity site means. Normal polarity directions are in situ, and reversed polarity means are inverted through the origin to facilitate comparison. Refer to Table A1 for the pertinent data.

(F±ΔF=11.9°±6.4°) is observed at the locality. A precision parameter value of 30 implies moderate scatter of site mean directions about the locality mean for adequate sampling of paleosecular variation. Lava flows and minor interbedded volcaniclastic rocks at the Garrett Rock locality strike NE and dip SE (Figure A2). The tilt correction is based on the mean of 42 measurements in the Garrett Rock area. The structurally corrected data may indicate 14.9°±10.5° of clockwise rotation since middle Eocene time.

Long Creek: Twelve autobrecciated flows were sampled at the Long Creek locality in a 500 m thick stratigraphic section (Figure 5d). The flows are lithologically identical to those at Garrett Rock, and similar demagnetization behavior supports this correlation (compare Figures 4c and 4e). Tilt-corrected inclinations at the two localities are similar and statistically indistinguishable (-52.0° and -47.5° at Garrett Rock and Long Creek, respectively). We were probably not successful at averaging secular variation at this locality. The flows are very uniform in composition, thickness and texture and may have been erupted during a short time interval as evidenced by the high value of the precision parameter k (83.5). All flows are reversely magnetized and the structurally corrected inclination is discordant (F±ΔF=16.5°±4.2°). The structurally corrected declination is almost 40° counterclockwise from the middle Eocene reference declination.

The orientation of flows at Long Creek (roughly N-S striking and steep dip to east) is very different from the attitude of similar flows at Garrett Rock, 8 km to the north. These disparate attitudes are most easily explained by about 45° of counterclockwise vertical axis rotation of flows at Long Creek relative to flows at Garrett Rock. Our paleomagnetic data appear to support such an interpretation. The tilt-corrected locality mean at Long Creek is rotated approximately 53° relative to that at Garrett Rock, roughly the amount indicated by the geological relationships.

Massacre Mountain: Lightning strikes were a significant problem at the Massacre Mountain locality but alternating field demagnetization generally allowed determinations of the characteristic magnetization direction (Figure 4f). Eight of fourteen sites from the Massacre Mountain locality required remagnetization circle analysis. Both normal and reversed polarity flows, tuffs, dikes, and sills were sampled. The site mean data pass a reversal test (Figure A1, Table A1). Directions from reversed polarity sites are more dispersed than directions from normal polarity sites (Figure 5e). The precision parameter associated with the locality mean is 18.2, and neither the rotation nor flattening are statistically significant (Table 2). The inclination flattening (F±ΔF=8.3°±8.4°) is at the limit of the associated confidence.

Meadow Creek: Sites at the Meadow Creek locality in the northern Lost River Range behaved well during alternating field demagnetization (Figures 5f and 4g). Site MC12 may be located within a small slump, and was therefore not included in the locality mean. Including this site mean would reduce the magnitude of the inclination flattening. The mean of the reversed polarity sites is included within the 95% confidence level about the mean of the normal polarity sites (Figure A1; Table A1). The structural data that constrain the tilt correction are shown in figure A1. The precision parameter associated with this locality (34.8) is somewhat high but is still within an acceptable range. About 30° of counterclockwise rotation and 9° of flattening are recorded at this locality (Table 2). Both the rotation and flattening are statistically significant at the 95% confidence limit.

Methodist Creek: The eight sites analyzed at Methodist Creek are reversely magnetized (Figures 5g and 4h). The lithologies in this section are quite variable and include dacite lavas, at the base, andesite lavas and interbedded tuffs, implying that significant time is represented within the section. The precision parameter of 26.3 suggests that the eight sites at this locality have reasonably sampled paleosecular variation. The structural correction at this

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Table A1. Average Directions of Normal and Reverse Polarity Sites

	Dec	Inc	R	k	α_{95}	N	Polarity
Copper Lake	292.5	34.2	7.7280	29.6	11.1	8	R*
	329.0	38.9	3.9896	391.3	5.4	4	N
Sawmill Canyon	300.5	27.2	6.5814	16.2	16.5	7	R*
	296.5	37.1	14.4450	27.0	7.8	15	N
Massacre Mountain	314.3	34.8	6.4337	12.4	19.4	7	R*
	312.5	37.0	4.9660	146.2	7.1	5	N
Meadow Creek	300.4	49.5	8.7561	36.9	9.1	9	R*
	287.6	59.6	1.9938	329.6	19.7	2	N

Dec, Inc, R, k, α_{95}, and N as in Table 1.
* Inverted through the origin.

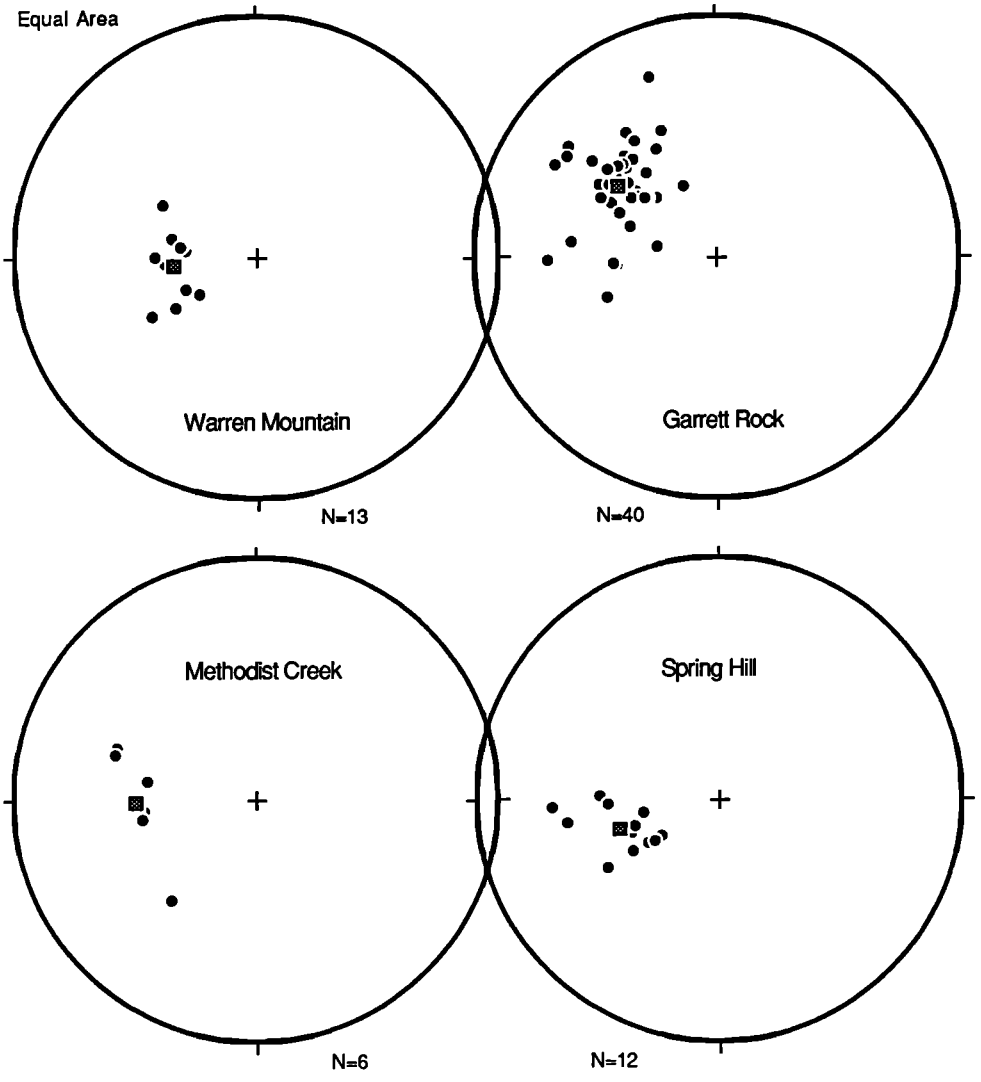


Fig. A2. Lower hemisphere equal area projection of structural data from Warren Mountain, Garrett Rock, Methodist Creek, Spring Hill, Copper Lake, Meadow Creek, and Sawmill Canyon. Shaded square denotes mean strike and dip. Meadow Creek data taken from Mapel et al. [1965]. Some of Sawmill Canyon data is from Ruppel and Lopez [1981].

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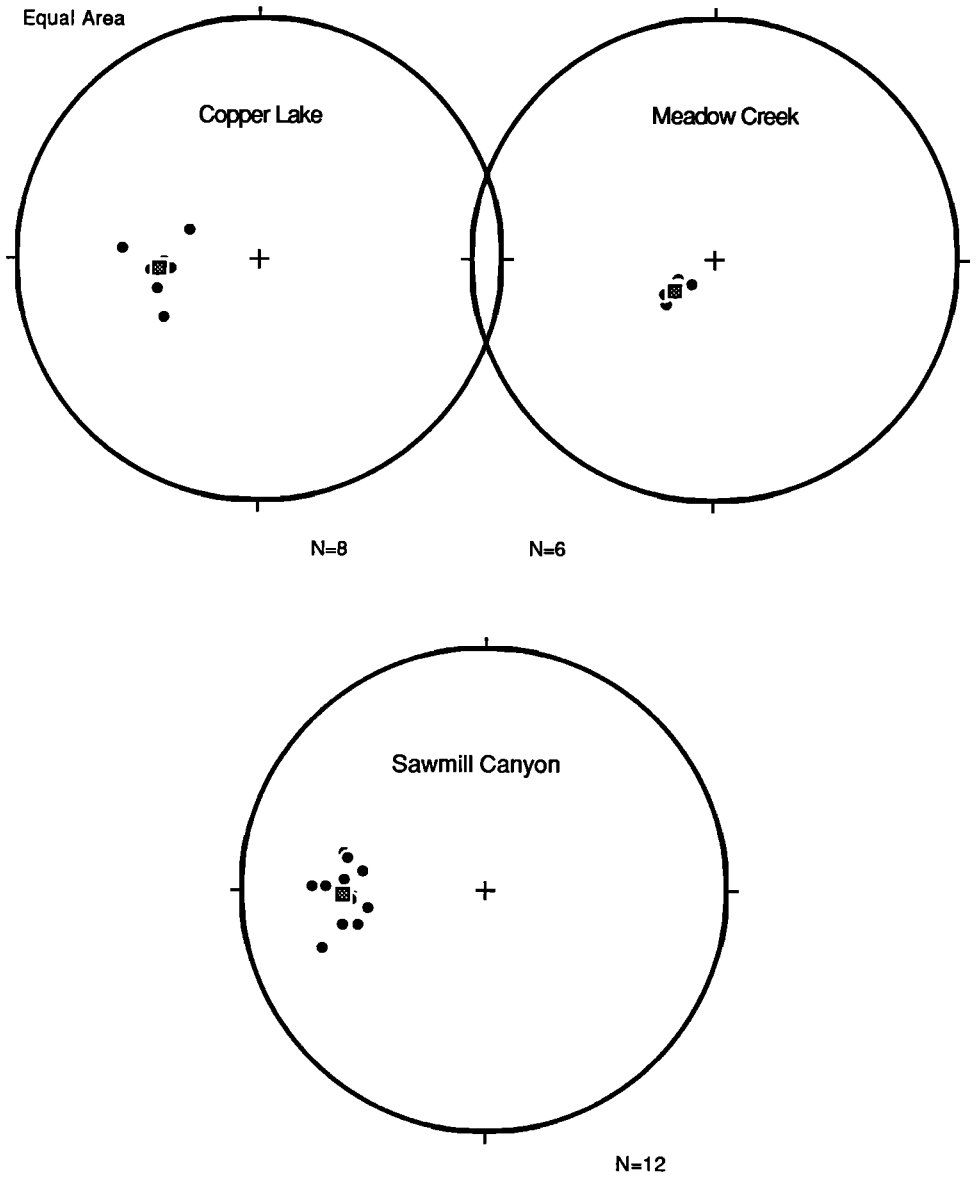


Fig. A2. (continued)

locality (Figure A2) is based on the attitude of tuffs interbedded with the sampled lava flows and on strikes and dips of overlying tuffs. The mean of six orientation measurements is 358.6°, 40.9° with an associated $\alpha 95$ of 16.2°. The structurally corrected locality mean direction does not differ at the 95% confidence level from the middle Eocene reference direction.

Sawmill Canyon: The Sawmill Canyon locality has the thickest and most complete section of Challis rocks in the area. Twenty-two intermediate lava flows and one ash flow tuff were sampled (Figure 5h). Three sites were also collected from indurated volcaniclastic conglomerates at the top of the section. The magnetization of the conglomerate samples was dominated by low-coercivity phases with unstable behavior. These data are not reported in Table 1. The flows are equally divided between normal and reverse polarity, and there are three polarity zones in the stratigraphic section (e.g., Figures 4i and 4j). According to the Berggren et al. [1985] magnetic polarity time scale, this section probably spans at least 3 m.y. from 49 to 46 Ma, and possibly as much as 4 m.y. The lower three quarters of the section is fairly uniform in lithology, consisting of interbedded intermediate plagioclase-hornblende-pyroxene-bearing lavas and tuff breccias. A biotite-bearing lava at the top of the section is overlain by poorly consolidated vitric tuffs and a normal polarity quartz-sanidine ash flow tuff (Figure 4j) which probably correlates with the 45-46 Ma tuff of Challis Creek [McIntyre et al., 1982].

The precision parameter of 20.0 suggests that paleosecular

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variation has been adequately sampled. Bingham statistics indicate that the distribution of site means about the locality mean is circular (Table 2; Figure 5). The mean of the normal polarity sites is included within the α_{95} of the mean for the reverse polarity sites (Figure A1; Table A1). The structurally corrected declination and inclination at Sawmill Canyon are concordant with the middle Eocene expected direction (R±ΔR=2.5±14.5°, F±ΔF=-1.9°±6.0°)(Table 2). These results strongly argue against significant vertical axis rotations at this locality.

Spring Hill: We collected more than 200 paleomagnetic cores from 25 sites near the top of the 2880 m high Spring Hill. Eight of the sites required remagnetization circle analysis, and three of these were subsequently excluded from the locality mean calculation because the calculated directions are far removed from the respective locality mean (Table 1). The remaining 22 flows are all reversely magnetized (e.g., Figures 4k, and 5i). The distribution of site mean virtual geomagnetic poles (VGP) about the locality mean VGP is circular, and the precision parameter of 25.1 suggests adequate sampling of paleosecular variation at the locality (Table 2; Figure 5i). The tilt-corrected remanence direction indicates significant counterclockwise rotation at the Spring Hill locality (R±ΔR= -43.0°±11.1°). The inclination is concordant with the expected direction (F±ΔF=2.3°±5.3°) (Table 2).

Warren Mountain: The Warren Mountain locality is situated a few kilometers north and west of the Methodist Creek locality and is composed of lava flows, tuff breccias, and a small plug at the base of the Challis volcanic section. The sequence is part of a dacite dome-flow complex. Lateral equivalents of the flows at Methodist Creek stratigraphically overlie the volcanic rocks at the Warren Mountain locality.

The demagnetization behavior of the samples was generally good (e.g., Figure 4l). Some samples at sites WM64 and WM66 at Warren Mountain exhibited exceptionally high median destructive fields, so demagnetization was completed thermally. All the flows at Warren Mountain are reversely magnetized (Figure 5j). The k value of 21.4 at Warren Mountain suggests that paleosecular variation may have been averaged at the locality. The site means are fairly evenly distributed about the locality mean. New ^{40}Ar-^{39}Ar plateau dates from site WM61 and a flow just above site WM66 show that more than 360,000 years were sampled by the volcanic rocks in the section (S.U. Janecke and L.W. Snee, unpublished data, 1989). Two sites were collected above the dated flows, and one additional site sampled an intrusive into flows below the dated flows. The estimated time period of 360,000 years should have been adequate to average westward drift and most of the dipole wobble of the geomagnetic field if volcanism was continuous during the interval. The lithologies of the volcanic rocks are quite uniform in this section and may have been erupted during short discrete time intervals, however, therefore paleosecular variation may not be adequately sampled.

The tilt correction is not as well determined at Warren Mountain as it is at most of the other localities. The base of the sequence rests upon a gently east dipping erosion surface. The sampled section consists of a monotonous sequence of plagioclase-hornblende-pyroxene-bearing lavas with locally well developed and extremely variably oriented flow banding. Towards the top of the sampled sequence, tuff breccias and poorly exposed ash flow tuffs alternate with intermediate lava flows. It was not possible to determine the attitude of flows in the lower two thirds of the sampled section. The tilt correction (Figure A2) is based mostly on measurements from lava flows above the paleomagnetic section and may not be representative of the entire locality. The same structural correction was not used at the Warren Mountain and Methodist Creek localities because there is good field evidence for differences in tilt at the two localities.

The structurally corrected remanence direction at Warren Mountain indicates 19.8° ± 16.5° of clockwise rotation and a statistically insignificant flattening of inclination (F±ΔF=8.2° ± 9.2°) (Table A1). The discordant declination is puzzling in light of the absence of rotation at the Methodist Creek locality a few kilometers to the south. Because of the uncertain tilt correction, we feel that the result from the Methodist Creek locality is a more realistic assessment of rotation than the result from the Warren Mountain locality.

APPENDIX 2

Possible errors and uncertainties in the paleomagnetic results are discussed below.

Inclination Flattening

Three locality means indicate statistically significant flattening of inclination, and four others, while not statistically significant, indicate flattenings greater than 5°. Possible explanations for the three statistically significant discordant inclinations include (1) an inclination bias in volcanic rocks, (2) incorrect or incomplete structural corrections, (3) insufficient averaging of paleosecular variation at some localities, (4) random noise [Calderone and Butler, 1988, 1991], (5) systematic errors in sample orientations or measurements; (6) incomplete removal of secondary components of magnetization, (7) long-term Eocene nondipole geomagnetic field behavior, (8) large-magnitude northward tectonic translation (roughly 1000 km), or (9) error in the reference pole. Option 8 can be excluded immediately because the study area is underlain by rocks of the Cordilleran miogeocline and is firmly attached to the North American continent. Clearly, northward translation since middle Eocene time can not explain the shallow inclinations. The quality of the reference pole from Diehl et al. [1983] suggests that option 9 is unlikely (see discussion in the text). We examine options 1, 2, and 3 below.

Inclination flattening of the order of 3° to 6° has been reported in Recent lava flows on the Hawaiian Islands [Castro and Brown, 1987]. The study in Hawaii found that Recent lava flows may not be perfectly accurate recorders of the ambient magnetic field. In addition, a compilation of numerous studies of prehistoric flows on Hawaii showed similar shallow discordant inclinations, 5.3° to 23.0° shallower than expected [Castro and Brown, 1987]. Discordant declinations were not observed in the Hawaiian data set. Kuntz et al. [1986] provide similar evidence in a study of latest Pleistocene to Holocene basalts in central Idaho. Examination of their Figure 2 indicates a bias of site means toward inclinations which are shallower than expected. Castro and Brown [1987] postulate that anomalous behavior in TRM acquisition may account for their observations. It is possible that some of the inclination bias detected in our data set is due to similar processes, but we have no way of demonstrating this.

Sensitivity of the Results to Uncertainties in the Tilt Correction

Apparent inclination and declination discordances may arise from incorrect structural corrections. For the data set from east central Idaho, in situ locality means and structural corrections are such that, in general, overestimating the dip of the

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sequence results in erroneous clockwise rotations and underestimating the dip results in erroneous counterclockwise rotations. The use of a strike for tilt correction which is counterclockwise from the true strike results in positive inclination anomalies, whereas strikes clockwise from the true strike yield inclination steepening (Figure A3).

The magnitude of this effect varies from locality to locality and may be considerable. For our data set a 10° error in determining the strike of the sequence can translate into a 4.5°-8° flattening of inclination, depending on the in situ direction at a particular locality. If incorrect structural corrections are the principal cause of the observed inclination flattening, then the strikes applied would have to be systematically biased about 10° in a counterclockwise fashion from the correct value. Random errors in strike estimates should give rise to flattened inclinations at some localities and steepened inclinations at others. A systematic bias in our application of strikes counterclockwise from the "true" value is unlikely because (1) the estimates of strike and dip are based on observations ranging from three-point solutions to direct measurements of strike and dip, (2) in many instances the strike of bedding is well defined on aerial photography, and (3) similar discordances were observed where the tilt correction relied on structural data collected during this study and where the tilt correction was derived from existing geologic maps.

Errors in estimating stratal dip result primarily in incorrect estimates of rotation (Figure A3). Apparent clockwise rotations of 20°-25° result from a 10° overestimate of the dip. About 15°-20° of apparent counterclockwise rotation result from a 10° underestimation of the dip. Dip is more difficult to constrain than strike in steeply tilted strata so small, random declination discordances, in the 5°-10° range, are possible. We feel that it is not possible to confidently discriminate between 5° and 10° of vertical axis rotation in tilted volcanic rocks because of this uncertainty, but vertical axis rotations exceeding 15° are readily resolved.

It is more difficult to define statistically significant vertical axis rotations paleomagnetically at high latitudes because of steep inclinations. In east central Idaho, vertical axis rotations can still be identified, but the resolution is poorer than at low latitudes. Vectors which differ only slightly in an absolute sense appear to differ significantly when one considers declination alone. Thus small errors in the tilt correction have a larger influence on apparent rotations at this latitude than similar errors at lower latitudes [Chan, 1988].

Field observations have defined three sets of normal faults which offset the Challis volcanic rocks in the study area. Two or more episodes of tilting about horizontal axes may yield apparent vertical axis rotations if the local resultant strike line is used for tilt correction (see discussion of this problem by MacDonald [1980]). This problem should be most acute in areas traversed by numerous nonparallel faults, such as the central Lost River Range where superposed NE, north, and NW striking normal fault sets segment numerous small crustal blocks (Figure 1). There is good evidence for tilting of Cenozoic strata along the N-S striking fault system. Tilting during latest Cenozoic Basin and Range as well as during Eocene NW-SE extension is also likely but the amounts of tilting caused by these two phases of extensional deformation are not clear. Geological mapping suggests that multiphase tilting is possible at Massacre Mountain, Spring Hill, Copper Lake, Long Creek, and Garrett Rock. A simple, single-phase tilting history is likely at Methodist Creek, Sawmill Canyon, Ellis, Meadow Creek, and Warren Mountain. The absolute amounts of tilt associated with each fault set are unknown. At every locality except Garrett Rock, volcanic rocks strike subparallel to a mapped fault in the area, suggesting tilting during a single faulting event. We do not believe that undetected multiple tilts have led to significant errors in our application of structural corrections.

Tilt Correction Analysis

In order to evaluate the reliability of our data at the locality level and to investigate possible initial dip of volcanic strata we solved for the structural tilt which restores each in situ locality mean to the expected direction (Table A2), assuming that each locality mean averages secular variation. We then considered whether these tilts are realistic given the structural geometry of each locality. If we can demonstrate that the required tilt is unrealistic, we have increased confidence that the inferred vertical axis rotation at that particular locality is a true tectonic rotation.

This analysis also allows detection of possible syntilting magnetizations or initial dips. If the required tilt correction differs from the observed tilt in dip but not strike, then initial dip is a likely explanation for the observation. For example, if volcanism was synchronous with extension, volcanic rocks might flow across inclined surfaces on hanging walls of normal faults and solidify with an initial dip toward the normal fault (Figure A4, sequence (A-B). After additional slip on the normal fault the apparent tilt (δ obs) of the lava flow would exceed the actual tectonic tilt (δ tectonic). In such cases, partial tilt correction (or undercorrection) would restore the paleomagnetic directions to their proper orientation. Bardoux and Irving [1989] present an example of this type of analysis.

Data in Table 4 show that overcorrection rather than undercorrection about a fixed strike line would reconcile observed and expected directions at the Meadow Creek, Spring Hill, and Long Creek localities. In each case, because the flows dip less steeply than predicted, initial dip directions of the flows would have been opposite in sense to the subsequent tectonic tilt (Figure A4, sequence B-C). Unless the volcanic rocks are deposited in the immediate footwall of a normal fault, this sequence of events is highly unlikely and is therefore rejected. We conclude that undetected initial tilt is not the cause of the large declination discordances at these localities.

For those localities where the standard treatment of the data suggests no statistically significant vertical axis rotation (Methodist Creek, Sawmill Canyon, Massacre Mountain, Copper Lake), the required tilt correction is within the range of acceptable tilts and is close to the preferred structural correction based on independent field data (Table A2).

Secular Variation

Middle Eocene time is characterized by relatively long magnetic polarity chrons with average duration greater than 1 m.y. [Berggren et al., 1985]. The flows in the study area probably erupted within C21N, C20R, and C20N. Most of the Challis volcanic rocks probably erupted during C20R, which is approximately 2.6 m.y. in duration. Thus it is not surprising that six of the 10 localities have only reversed polarity flows (Table 2). We argue that the absence of reversals in most of our sections does not necessarily imply insufficient averaging of paleosecular variation.

Site mean directions from reverse polarity sites are more scattered than those from normal polarity sites at all the dual polarity localities in this study. This pattern could arise if all normal polarity flows were erupted over single, short time

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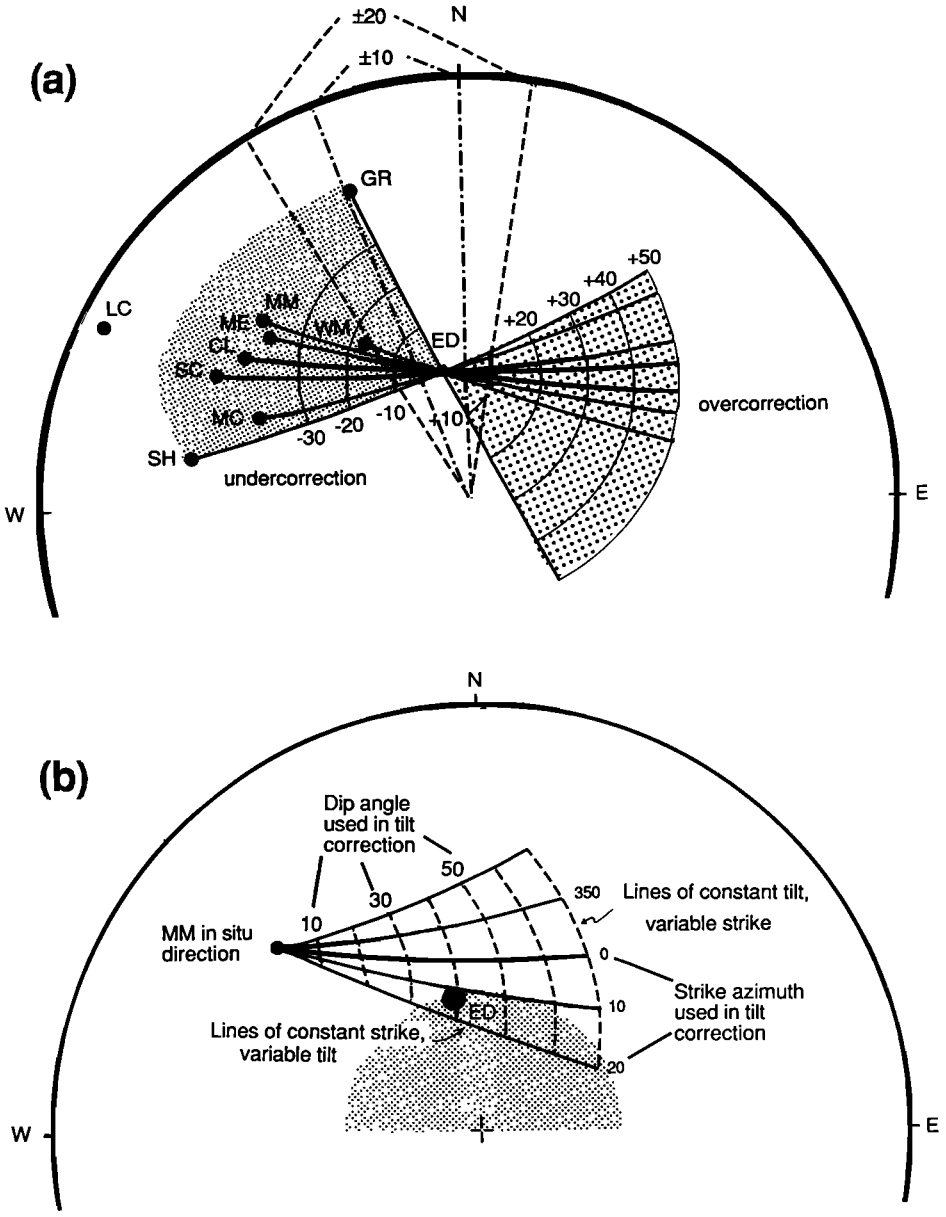


Fig. A3. (a) Lower hemisphere equal area stereonet showing apparent vertical axis rotations which result from over- and underestimating the dip of volcanic sequences. In this construction we assume that the in situ observed directions arise solely from tilting about horizontal axes. Overcorrecting (using a dip steeper than the true dip) results in apparent clockwise rotations, whereas undercorrecting results in apparent counterclockwise rotation at every locality except the Garrett Rock locality. Small circle trajectories connecting the expected direction and in situ directions are incremented progressively 10° in dip from the expected direction (ED). Shading denotes zones of over- and undercorrection. Implied vertical axis rotations of ±20° lie between the dashed lines, whereas rotations of ±10° lie between the dot-dashed lines. Locality means are: CL, Copper Lake; GR, Garrett Rock; LC, Long Creek; MM , Massacre Mountain; MC, Meadow Creek; ME, Methodist Creek; SC, Sawmill Canyon; SH, Spring Hill; and WM, Warren Mountain. (b) Lower hemisphere equal area stereonet showing trajectories of the observed in situ direction at the Massacre Mountain locality (as an example) being corrected for easterly dip about various strike azimuths. From these trajectories it is seen that use of strike lines between azimuth 340° and 10° produce inclination flattening in the structurally corrected direction. The region with inclinations steeper than expected is shaded.

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Table A2. Initial Dip Test

Locality	Inverted Strike	Inverted Dip	Realistic Attitude ?	Actual Strike	Actual Dip
Copper Lake	5.5	43.0	Yes	354.0	34.0
Garrett Rock	63.0	41.5	No. Strike off >20°	36.5	41.3
Long Creek	4.0	78.0	No. Strike correct, dip too high	0.0	50.0
Massacre Mountain	14.5	40.0	Possibly, strike somewhat off	0.0	45.0
Meadow Creek	347.0	33.5	No. Strike correct, dip too steep	320.2	16.6
Methodist Creek	12.0	41.0	Yes	358.6	40.9
Sawmill Canyon	358.0	48.5	Yes	358.0	48.6
Spring Hill	339.0	56.0	No. Strike correct, dip too high	342.8	35.1
Warren Mountain	24.0	17.5	No. Strike off, dip seems too low	354.0	28.3

Strikes and dips are in degrees.

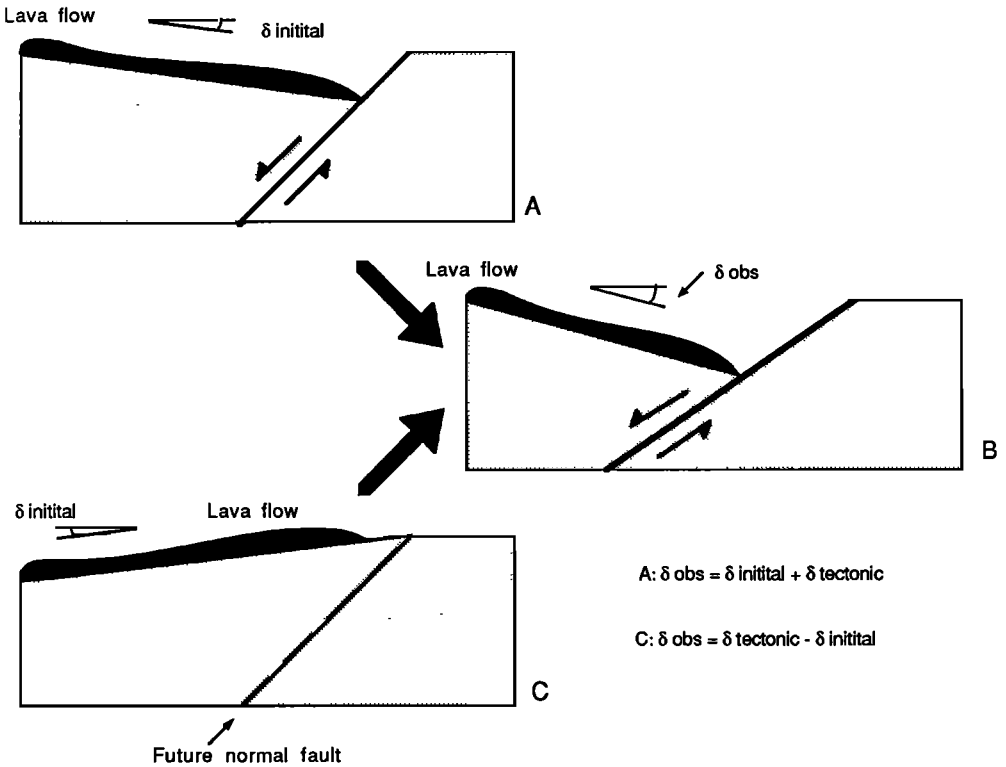


Fig. A4. Schematic cross sections showing origin of initial dip of volcanic rocks in the hanging wall of a normal fault. A lava flow is deposited across an inclined surface in cross sections A and C. Sequence A-B depicts the typical situation where tectonic tilting is less than the apparent dip of the lava flow (δ_{obs}). Sequence C-B depicts the unlikely alternative where the tectonic tilt is greater than the final apparent dip of the flow. In this case the initial dip of the flow would have to be opposite in sense to the subsequent tectonic tilting.

intervals. If this were the case, our analysis would underestimate the dispersion in the data and overestimate the number of independent directions. We discount this hypothesis because normal polarity flows are just as heterogeneous in lithology as reversed polarity flows and normal and reversed polarity flows are lithologically indistinguishable from one another in the field. K-Ar ages from sections of lava flows at the base of the Challis Volcanic Group in central Idaho indicate that normal polarity flows were erupted during a 2 m.y. period between approximately 51 and

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49 Ma [McIntyre et al.; 1982; Armstrong, 1975; Fisher et al. 1983].

Mankinen [1989], Sheriff [1989], and Calderone et al. [1990] recently have shown that it is commonly difficult to average secular variation with volcanic sequences, in part because of the episodic nature of volcanism. Both Mankinen [1989] and Sheriff [1989] observed statistically significant flattening of inclinations in the Sonoma and Hog Heaven volcanic fields, respectively. Mankinen [1989] argued that the elliptical distribution of virtual geomagnetic poles derived from sets of lava flows may indicate insufficient averaging of paleosecular variation. Distributions of virtual geomagnetic poles from all but two of the localities (Spring Hill and Sawmill Canyon) are elliptical and have axial ratios above 1.5 (α_{31}/α_{12} ratio of approximate 95% confidence angles determined using Bingham statistics [Onstott, 1980] (Table 2). This may indicate insufficient averaging of paleosecular variation. Apparent elliptical distributions of VGPs may be an artifact of the small number of site means at some of the localities as well as distortions produced by one or two outliers. While it could be argued that only the Spring Hill and Sawmill Canyon localities provide adequate averages of paleosecular variation, we find low to moderate precision parameters and the consistent relationships between strike and paleomagnetically determined vertical axis rotations (Figure 6; see discussion in text) to be quite compelling and believe that most of the locality means provide a reasonable average of paleosecular variation.

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